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Light interactions, use and efficiency in row crop canopies under optimal growth conditions

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ABSTRACT

Accurate estimates of light (Photosynthetically Active Radiation or PAR) absorption, in addition to other interactions is imperative to quantify growth, productivity, energy and water balance and other physiological and biophysical processes in any vegetative surface. Currently, a comparative assessment of light interaction patterns across row crops is lacking, especially under current levels of productivity achieved in the U.S. High Plains. Here, we continuously measured canopy light balance components at high-frequency (15 min) to characterize transmittance (R), reflectance (R), fraction of intercepted PAR (fIPAR), fraction of absorbed PAR (fAPAR), light extinction coefficients (k), and light use efficiency (LUE) comparatively across maize, soybean, sorghum and winter wheat under optimal growth conditions. While maximum fAPAR was 88-96% of incoming PAR in all crops, mean fAPAR varied from 82% in sorghum to 46% in winter wheat; while k ranged from 0.36 (winter wheat) to 0.48 (sorghum and soybean), and these differences reflect highly crop-specific signatures. Conversion factors among fIPAR and fAPAR and LUE based on either component (LUE_i and LUE_a) were quantified that were substantially different from the conventionally used values; especially during early and late growth stages. A commonly employed approach of solar-noon light sampling was evaluated, and it was found that early and late stages of crop growth experience greater potential errors (as high as 110%) under this sampling approach, and hence should be avoided. LUE_a was the highest for maize (5.3 g MJ^{-1}), followed by sorghum (4.1 g MJ^{-1}), winter wheat (4.0 g MJ^{-1}) and soybean (3.1 g MJ^{-1}). The datasets measured, analyzed and interpreted here present unprecedented quantities of biomass productivity and canopy light use parameters for four major cropping systems, and hence, should be accounted for in crop growth and productivity modeling applications to enhance predictive accuracy and robustness.

1. Introduction

Quantification of crop growth and its drivers has long been a subject of investigation because of their critical importance for a number of applications, including modeling of crop growth, yield, and response to environmental factors (Kukal and Irmak, 2018a,b). Thus, the robustness of outcomes of these modeling exercises is a function of how accurately the underlying biophysical parameters in crop models are defined. Crop growth has been chronologically expressed as a function of time (Blackman, 1919), progressed to incorporate leaf assimilatory area (Watson, 1952), and has theoretically and experimentally been shown to be determined by intercepted solar radiation (Photosynthetically Active Radiation, referred to "light"/"PAR" hereon) (DeWit, 1959; Loomis and Williams, 1963). Under the accepted status quo, potential biomass accumulation is governed by the proportion of incident PAR that is intercepted if no biotic/abiotic stresses exist (Monteith, 1972, 1977; Kiniry et al., 1989; Sinclair and Muchow, 1999). Intercepted PAR is a function of leaf area index (LAI) and crop architecture (Russell et al., 1989; Guiducciet al., 1992). The incident PAR (prior to interaction with vegetation or soil) depends on geographical location, weather patterns, and time of year.

Related to canopy light interception, the light use efficiency (LUE) concept has crucial applications to model crop growth and performance, owing to the use of effective parameters that can be measured relatively easily (Brisson et al., 2003; Muchow et al., 1990; Jones et al., 1986). Crop models are based on "growth engines" that can be "carbon-driven", "water-driven", or "solar-driven", and use solar energy in differing structures, emphasis and parameterizations (Steduto, 2003; Albrizio and Steduto, 2005). Some of the major crop models that rely on LUE concept are CERES: Crop Estimation through Resource and Environment Synthesis (Jones et al., 1986), EPIC (Jones et al., 1991), STICS: Simulateur mulTIdisciplinaire pour les Cultures Standard

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(Brisson et al., 2003) and CROPSYST: Cropping Systems Simulation Model (Stockle et al., 2003). The empirical/measured LUE can be multiplied with intercepted (or absorbed) PAR to calculate daily gross dry matter production (Lecouer and Ney, 2003). Several parameters involved in these processes, including LAI, light extinction coefficient (k), intercepted and absorbed PAR, and LUE, have to be measured/estimated with adequate accuracy and precision, for successful modeling of crop growth and related functions.

A wide array of crops' light interception and LUE has been studied, under contrasting environments and management regimes. Some examples, listed as crop studied and geographical location of the research, are maize (Iowa, USA: Hatfield 2014;Singer et al. 2011; New Zealand: Teixeira et al. 2014; France: Plenet et al. 2000; Netherlands: Gou et al. 2017; Canada: Rochette et al. 1996; Argentina: Maddoni and Otegui 1996; Arkansas, USA: Edwards et al. 2005; Nebraska, USA: Lindquist et al. 2005; China: Gao et al. 2010;Wang et al. 2010), soybean (Missouri, USA: Van Roekel and Purcell 2014; Iowa, USA: Hatfield 2014;Singer et al. 2011); Arkansas, USA: Purcell et al. 2002), sorghum (Kansas, USA: Narayanan et al. 2013; Italy: Steduto and Albrizio 2005; Ceotto et al. 2013; New Zealand: Fletcher et al. 2013), wheat (Italy: Steduto and Albrizio 2005; Netherlands: Gou et al. 2017; Argentina: Caviglia and Sadras 2001; China: Li et al. 2008; Wang et al. 2010; Australia: O'Connell et al. 2004; New Zealand: Fletcher et al. 2013), rice (Texas, USA: Campbell et al. 2001), barley (Washington, USA: Kemanian et al. 2004); UK: Bingham et al. 2007), sunflower (Italy: Steduto and Albrizio 2005), chickpea (Italy: Steduto and Albrizio 2005), giant reed (Italy: Ceotto et al. 2013), bell-pepper (Portugal: Vieira et al. 2009), mustard (India: Pradhan et al. 2014; Australia: O'Connell et al. 2004), lettuce (UK: Tei et al. 1996), onion (UK: Tei et al. 1996), red beet (UK: Tei et al. 1996), potato Spain: Camargo et al. 2016), field pea (Australia: O'Connell et al. 2004), quinoa (Chile: Ruiz and Bertero 2008); peanut (Texas, USA: Kiniry et al. 2005), forage rape (New Zealand: Fletcher et al. 2013), olive (Spain: Mariscal et al. 2000). Several studies have also conducted meta-analysis of crop LUE from different environments Kiniry et al., 1989; Sinclair and Muchow, 1999). Although, a reasonable quantity of research is based on the U.S. Great Plains and its constituent cropping systems, the region lacks: ((1) a comparative assessment of light interactions patterns in different crop canopies, (2) implementation of a complete light balance, and (3) quantification of LUE for the major crops in this region.

The selection of crops to be included in this research was justifiable given several factors. Firstly, these are the four major row crops in Nebraska accounting for 86% of the total harvested Nebraska cropland (USDA-NASS, 2018) and in the U.S. Great Plains, in general. Fig. 1 presents the spatial distribution of cropland that was grown in 2018 within and around Nebraska (USDA Cropscape, 2018). Secondly, these crops have a substantial share of irrigated cropland in the state, which is evident in Fig. 1 from the overlap of center-pivots and the cropland under these crops, although a considerable land area under gravity (furrow) irrigation also exists in the state. Hence, these crops were grown under optimally irrigated conditions so as to allow them to reach their highest production potential. Thirdly, these crops also provide immense opportunity to identify and study differences in light balance dynamics over these canopies as they differ in several aspects. These include their mechanism of photosynthetic pathways (C_3, C_4), canopy architecture (spherical, heliotropic), leaf angle distribution (erectophile, planophile), ground cover fraction, row architecture (row distance, plant-to-plant distance etc.), leaf morphology, resource use (water, light, nitrogen), plant densities etc. With these multifaceted differences, these crops are ideal for an analysis that aims to differentiate vegetation-light interaction patterns in a quantitative manner under homogenous growing conditions.

The evaluation of various crops' light interaction and LUE dynamics in different environments has been performed with contrasting approaches, which could possibly impact LUE estimates. This research has made efforts to recognize and minimize these approach-driven impacts and quantify robust characteristics of canopy-light interactions. Firstly, we address the tradeoffs among the two light sampling approaches commonly adopted in research investigations. While majority of the research estimates LUE by measuring intercepted/absorbed light (or their proportion in incident light, i.e., fIPAR/fAPAR) using several instantaneous sampling events during the growing season, it is ideal to sample these variables at high-frequency (sub-hourly basis, preferably) throughout the growing season. Although instantaneous measurements in these studies are conducted around solar noon, they might present a non-representative condition of crop canopy vs. light interactions. This is due to the dynamic nature of leaf angle and transient cloud and wind conditions, that can alter the typical inter-row ground shadow signature at a given crop phenological stage during the day. Thus, scaling the instantaneously-measured fIPAR/fAPAR to represent the entire day seems implausible, especially in environments like Nebraska, characterized by transient cloud cover, high wind speeds, abrupt changes in weather conditions, etc. Moreover, semi-arid environments (eg. South-Central Nebraska) can also see temporary morphological changes such as leaf curling in maize, and wilt response in soybean leaves during heat stress events (high vapor pressure deficit), despite optimal growing conditions and adequate soil moisture. These morphological responses alter the light interactions from temporarily to long enough durations to be sufficient to cause erroneous light interception data. A continuous light measurement approach has been adopted by research (Lindquist et al., 2005) in recent times with optical technology becoming assessible and affordable (Salter et al., 2019). Interpolating fIPAR/fAPAR magnitudes for the intervals between sampling dates assumes that crop growth is a linear function of time. Although, this approach has been long outdated (Sinclair and Muchow, 1999), LUE research nevertheless, depends on time as the independent variable to model crop light interception/absorption characteristics. We argue that in situations where light measurements cannot be continuous and high-frequency from emergence to harvesting, more relevant drivers such as thermal units (or growing degree days) can be used as proxies to model fIPAR/fAPAR. Although short-term variability in light interactions can arise from transient wind and cloud conditions, thermal units is an effective driver of crop development, and hence canopy light interactions (Kukal and Irmak, 2019). Secondly, a large proportion of studies rely on intercepted light to estimate LUE, when actually absorbed light represents what is available for photosynthesis and plant metabolic functions. This presents confounding situations when LUE values are compared across literature, if the basis (fIPAR vs. fAPAR) for LUE estimation in the research are not considered (LUE based on fIPAR tend is lower than that on FAPAR). The relationship of fIPAR and fAPAR is often subject to sub-seasonal variation, which makes conversion of LUE based on one to the other is challenging (Bonhomme, 2000; Sinclair and Muchow, 1999). Thirdly, the non-stationarity of canopy light interactions, especially light extinction coefficient (k) and LUE for a given crop canopy is often neglected. k is extensively used as an input in crop modeling applications, and also assumed to be constant during crop growth during the growing season, equal to an empirical value. For example, Irmak and Mutibwa (2008) showed that k not only changed during the growing season for maize canopy, but also fluctuated significantly within a day due to change in the sun angle and other factors. In this research, we propose that sub-seasonal variability in k can be measured and potentially modeled using efficiently transferable indicators, so that the use of a constant k can be avoided. This research took into consideration all of the above-mentioned gaps and limitations, so that a wide array of indicators employed to characterize vegetation-light interactions can be quantified accurately.

The specific objectives of this research were to: (i) characterize maize, soybean, sorghum and winter wheat crops for their light (PAR) interactions via quantifying components of a light balance in each canopy growing under optimal growing conditions; (ii) compare continuous and instantaneous light balance measurement approaches to

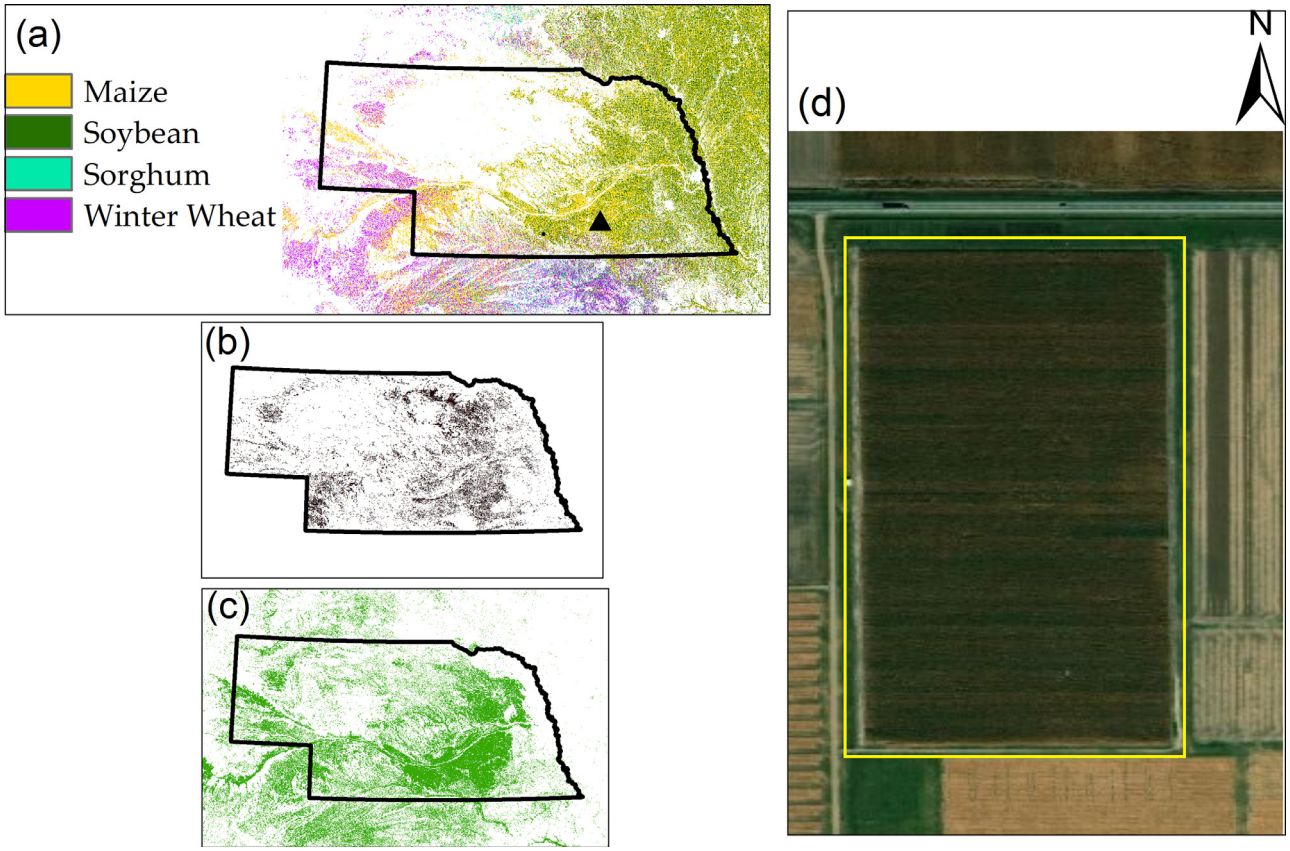


Fig. 1. Map showing (a) distribution of cropland in and around Nebraska where maize, sorghum, soybean and winter wheat was planted in 2018, (b) distribution of center pivot irrigation systems in Nebraska, (c) distribution of irrigated cropland in and around Nebraska, and (d) aerial view of the 13.5 ha subsurface drip-irrigated experimental field used in the research.

efficiently characterize crop canopies; (iii) quantify PAR-based albedo and light extinction coefficients during the growing season for each crop canopies; (iv) quantify and study seasonal and sub-seasonal patterns of LUE based on both intercepted and absorbed light for each crop.

2. Material and Methods

2.1. Research Site characteristics

The field investigations were carried out at the University of Nebraska-Lincoln, South-Central Agricultural Laboratory (40.58 N, 98.13 W; 552 m above mean sea level) near Clay Center, Nebraska. The soil at the site is a Hastings silt loam, well-drained upland soil (fine, montmorillonitic, mesic Udic Argiustoll) with 0.34 m^3 m^{-3} field capacity, 0.14 m^3 m^{-3} permanent wilting point, and 0.53 m^3 m^{-3} saturation point (Irmak, 2010). The total available water holding capacity of the soil profile is 240 mm 1.20 m^{-1}. The particle size distribution is 15% sand, 65% silt, and 20% clay, with 2.5% organic matter content in the topsoil (Irmak, 2010). The 13.5-hectare experimental field (Fig. 1d) was irrigated using a subsurface drip irrigation (SDI) system that was installed in 2004 by Dr. Suat Irmak and Ross Roberts (Diversity D, Inc., Lubbock, Tex.) and is the first SDI field in Nebraska and is one of the largest experimental SDI fields in the U.S. (Irmak and Djaman, 2016). The long-term average annual rainfall in the area is 680 mm, with significant annual and growing season variability in both magnitude and timing. For detailed information on meteorological conditions at the research site, refer to Kukal and Irmak (2019) and Kukal and Irmak (2020).

2.2. Soil and crop management

The field experiments were conducted for the two growing seasons, i.e., 2016 and 2017 for maize (*Zea mays* L.), soybean [*Glycine max* (L.) *Merr.*] and sorghum [*Sorghum bicolor* (L.) *Moench*], and 2016-2017, and 2017-2018 for winter wheat (*Triticum aestivum* L.). For maize, a short season variety (80-day hybrid) was selected during both years, and a longer season variety (113-day hybrid) was included in 2017. To accommodate all these crops within the extent of the experimental field, the field was divided in the N-S direction into smaller independent plots, each dedicated to a single crop grown in E-W rows. Subdivisions within the uniform soil type ensured that all the crops experienced homogenous soil conditions. During each growing season, care was taken that where possible, crop management was uniform across all crops, while still representing the regionally prevalent crop management practices. All crops were fertilized appropriately and sufficiently, and the nitrogen amount applied to each crop was based on the soil samples taken from several locations in each cropped area of the field and the University of Nebraska-Lincoln nitrogen recommendation algorithms. Herbicide, insecticide and fungicide applications were made to all crops uniformly when needed. To maintain optimum growth conditions, it was taken care for that none of crops were water stressed. This was done by continuously monitoring soil moisture and applying irrigation each time the soil-water in the root zone for any of the crops was depleted by about 40–45% of the total soil-water. In addition to nutrients and water, the experiment was intensively managed to ensure avoidance of any stresses from weeds, insects and diseases. Table 1 lists various detailed agronomic and management information for each of the crops during the two growing seasons.

Table 1
Detailed management and agronomic information for the field experiment.

Detail	Unit								
Crop		S.S. Maize		L.S. Maize	Sorghum		Soybean		Winter Wheat	
Growing season		2016	2017	2017	2016	2017	2016	2017	2016-17	2017-18
Cultivar		DKC 30-19 RIB	DKC 30-19 RIB	Dekalb DKC63-60RIB	DKS 44-20	DKS 44-20	AG 2431	AG 2431	SY Wolf	SY Wolf
Sowing density	seeds ha^{-1}, kg ha^{-1} for winter wheat	84,767	84,767	84,767	280,939	280,939	371,654	371,654	101	101
Actual Plant Density	plants m^{-2}	8.5	8.5	8.5	28.1	28.1	37.2	37.2	278.8	278.8
Planting depth	cm	6.4	6.4	6.4	3.8	3.8	3.2	3.2	1.3	1.3
Row distance	cm	76.2	76.2	76.2	76.2	76.2	76.2	76.2	17.8	17.8
Row direction		E-W	E-W	E-W	E-W	E-W	E-W	E-W	E-W	E-W
Sowing	DOY	116	114	114	134	129	134	129	273	290
Emergence	DOY	132	133	130	146	143	144	136	281	301
Harvest	DOY	231	264	292	277	293	272	285	198	192
Max CTU (since sowing)	°C	1291	1570	1726	1658	1688	1627	1660	2920	2305
Max CTU (since emergence)	°C	1231	1503	1671	1606	1609	1596	1608	2803	2179
Irrigation amount	mm	254.00	279.40	279.40	285.75	279.40	285.75	279.40	152.40	190.50
Precipitation received	mm	331	407	538	341	467	341	467	385	282
Fertilizer Management	Date	3/14/2016 4/23/2016	2/9/2017 4/19/2017	2/9/2017 4/19/2017	3/14/2016 4/23/2016	2/9/2017 4/25/2017	3/14/2016	2/9/2017	2/9/2017 4/13/2017	1/29/2018 4/24/2018
	Type	11-52-0 32-0-0	11-52-0 32-0-0	11-52-0 32-0-0	11-52-0 32-0-0	11-52-0 32-0-0	11-52-0	11-52-0	11-52-0 32-0-0	11-52-0 32-0-0
	Method	Broadcast In furrow	Broadcast In furrow	Broadcast In furrow	Broadcast In furrow	Broadcast In furrow	Broadcast	Broadcast	Broadcast Liquid broadcast	Broadcast Liquid broadcast
	Amount	122 kg ha^{-1} 196 kg ha^{-1}	224 kg ha^{-1} 146 kg ha^{-1}	224 kg ha^{-1} 173 kg ha^{-1}	122 kg ha^{-1} 140 kg ha^{-1}	224 kg ha^{-1} 173 kg ha^{-1}	122 kg ha^{-1}	224 kg ha^{-1}	224 kg ha^{-1} 67 kg ha^{-1}	224 kg ha^{-1} 67 kg ha^{-1}
Weed Management	Date	5/3/2016	4/8/2017 5/6/2017	4/8/2017 5/6/2017	5/18/2016	4/18/2017 5/24/2017	5/18/2016	5/11/2017 4/8/2017	9/20/2016 4/18/2017	
	Type	Verdict + Roundup	2-4-D Ester + Roundup Verdict + Roundup	2-4-D Ester + Roundup Verdict + Roundup	Lexar + Roundup	2-4-D Ester + Roundup Warrant + Aatrex 4L	Valor + Roundup	Valor XLT + Roundup 2-4-D Ester + Roundup	Liberty 2-4-D	
	Amount	1.1 L ha^{-1} + 2.3 L ha^{-1}	0.9 L ha^{-1} + 2.3 L ha^{-1} 1.1 L ha^{-1} + 2.3 L ha^{-1}	0.9 L ha^{-1} + 2.3 L ha^{-1} 1.1 L ha^{-1} + 2.3 L ha^{-1}	7 L ha^{-1} + 2.3 L ha^{-1}	0.9 L ha^{-1} + 2.3 L ha^{-1} 4.7 L ha^{-1} + 2.3 L ha^{-1}	0.3 L ha^{-1} + 2.3 L ha^{-1}	0.4 L ha^{-1} + 3.5 L ha^{-1} 0.9 L ha^{-1} + 2.3 L ha^{-1}	2.1 L ha^{-1} 0.9 L ha^{-1}	
Light Balance monitoring initiated	DOY	160	133	130	172	143	172	151	281	301
Light Balance monitoring concluded	DOY	231	264	292	277	293	272	285	198	192

2.3. Field measurements

2.3.1. Aboveground dry matter

Every 1-1.5 weeks during both growing seasons, four quadrats of area with 1 m^2 each were destructively sampled (Kukal and Irmak, 2019). This sampling campaign was conducted for each crop, totaling to 16 samples (every 1-1.5 weeks) across all four crops (4 crops × 4 samples) in 2016 (or 2016-2017 for winter wheat) growing season, and 20 samples (every 1-1.5 weeks) across all crops (5 crops × 4 samples) in 2017 (or 2017-2018 for winter wheat) growing season. Each harvested area was at least 4-5 meters from areas harvested in previous campaigns, so that the harvested gaps do not contribute to any edge effects in areas to be harvested in the future. The destructive sampling was also limited to a particular (but representative) section of the field, in order to prevent random gaps in the field, which could have hindered the crop yield assessments from the records of the yield monitor. The samples were dried in a walk-in oven at 60°C until constant weight was attained and following which, dry matter from all the samples was weighed and recorded. We will refer to this aboveground biomass/dry matter as AGB hereon.

2.3.2. Leaf area index

Leaf area Index (LAI) was recorded using an AccuPAR LP-80 ceptometer (METER Group, Pullman, WA, USA) every 1-1.5 weeks at 20 random (and representative) locations in each crop canopy. The Ceptometer consists of a light bar with 80-point quantum flux sensors, which aid in detailed sampling and spatial averaging of light (Photosynthetic Photon Flux Density, or PPFD) incident on a surface. Since the light transmission into a row crop canopy is subject to high heterogeneity, owing to non-uniform patterns of direct and diffused light transmitted to the ground surface, a ceptometer is an ideal instrument to measure light transmission or LAI in row crop canopies. The instrument requires a light (PPFD) measurement to be recorded above the canopy, uninterrupted by any plant part to represent the incident light on the canopy, and several (flexible number of) measurements at the ground level to represent the transmitted light (PPFD) in the inter-row spacing. The LAI measurements were only conducted on clear sky days, with non-overcast or non-cloudy conditions, to maintain consistent measurement conditions throughout the growing seasons (Kukal and Irmak, 2019). LAI has been shown to have a diurnal variation owing to changes in solar angles (Weiss et al., 2004); hence it was necessary to measure LAI at the same time (around solar noon) every week. In this research, LAI refers to the total leaf area index measured in the field conditions.

2.3.2. Light (PAR) measurements

Two different approaches were used to measure light interactions in each crop canopy. These approaches differed in their nature of data collection frequency. The following sections describe these two approaches in detail.

Continuous light measurement setup. Under this approach, light interactions were measured using permanently installed sensors (Fig. 2) in each crop canopy. All the variables were observed every minute and recorded every 15 min as averages continuously during the entire growing season. In each crop, identical instrumentation was used. Three variables were observed under this approach:

(a) Incoming PAR flux (PAR_{in}): This is the uninterrupted incoming PAR that is received at the Earth's surface. This was measured using a point quantum sensor (Apogee Instruments Inc., Logan, UT, USA), which measures PAR portion of the electromagnetic spectrum (400-700 nm) and was mounted at the top of the canopy. Since the incoming PAR is not a function of vegetative surface, this measurement was made at a single point at the research field.
(b) Transmitted PAR flux (PAR_{tr}): This is a time-variant portion of the PAR_{in} that gets transmitted through the canopy and reaches the ground (soil). Since, this component is subject to spatial variability in the inter-row spacing because of the complex shading effects of the canopy, using a point quantum sensor would have resulted in erroneous and non-representative measurements. Thus, a line quantum sensor was mounted at the ground level (5-10 cm above soil surface) in the inter-row spacing in a diagonal orientation. A line quantum sensor (Apogee Instruments Inc., Logan, UT, USA) spatially averages the outputs of 6 individual PAR sensors placed linearly across the inter-row space. Each crop canopy was subjected to independent measurement of PAR_{tr}, unlike PAR_{in}.
(c) Reflected PAR flux (PAR_{ref}): This is a time-variant portion of PAR_{in} that gets reflected from the canopy and/or the soil back to the surroundings. This component is also subject to spatial variation because of canopy non-uniformity, and hence, a line quantum sensor was used in an inverted fashion, aligned to monitor the top of the canopy. Similar to PAR_{tr}, PAR_{ref} was measured independently in each crop canopy.

These variables were measured in units of quantum flux or PPFD (μmol m^{-1} s^{-1}), and the conversion of quantum flux to units of energy require detailed solar spectral data, which is expensive to collect. However, a conversion value of 4.57 μmol J^{-1} was used, which is a conservative estimate under diverse atmospheric conditions, as it is centered at 550 nm across the PAR range (McCree, 1972; Jones et al., 2003; McCartney 1978; Jacovides et al., 1997; Tollenaar and Aguilera, 1992; Lindquist et al., 2005). All the sensors were frequently maintained to be clean from any unwanted foreign material covering the optical sensor such as soil particles, plant material, residue etc.

Instantaneous light measurements. The same three PAR variables were also sampled instantaneously at frequent intervals (1-1.5 weeks) using AccuPAR LP-80 ceptometer. In each crop canopy, the PAR_{in}, PAR_{tr} and PAR_{ref} were measured by holding the ceptometer above the canopy, below the canopy (just above the soil surface) and inverting the light bar over the canopy to capture the upwelling light, respectively. These measurements were conducted at 20 locations in each crop. Prior to measurements, the ceptometer was calibrated manually using a manufacturer's calibrated point quantum flux sensor that was mounted and levelled on the ceptometer. The ceptometer measurements were also in quantum flux units and were converted to energy units, using the conversion factor mentioned earlier. These measurements, although being fixed in time, served to inform us about the standard deviation (spatial variability) associated with light flux, which could not be acquired from the continuous light balance measurements setup.

2.3.3. Light Balance

The fate of PAR_{in} can take three possible outcomes, depending on the surface characteristics. These possible outcomes, described in Eq. (1), when summed, should be equal to PAR_{in}, similar to a mass or energy balance:

$$PAR_{in} = PAR_{tr} + PAR_{abs} + PAR_{ref} \quad (1)$$

where, PAR_{in}, PAR_{tr}, and PAR_{ref} have been described earlier, and PAR_{abs} is the quantity of light absorbed by the canopy to be used in photosynthesis. This component of the light balance is also time-variant. It is calculated as a residual from Eq. (1) when all the other components are measured using approaches outlined in 2.3.2.1 and 2.3.2.2.

2.3.4. Indices describing light interaction

Transmittance and reflectance. Transmittance (T) and reflectance (R) are two indices that characterize a vegetative surface using the relative magnitudes of two measured components of light balance to the total light available for different interactions. Eqs. (2) and (3) provide

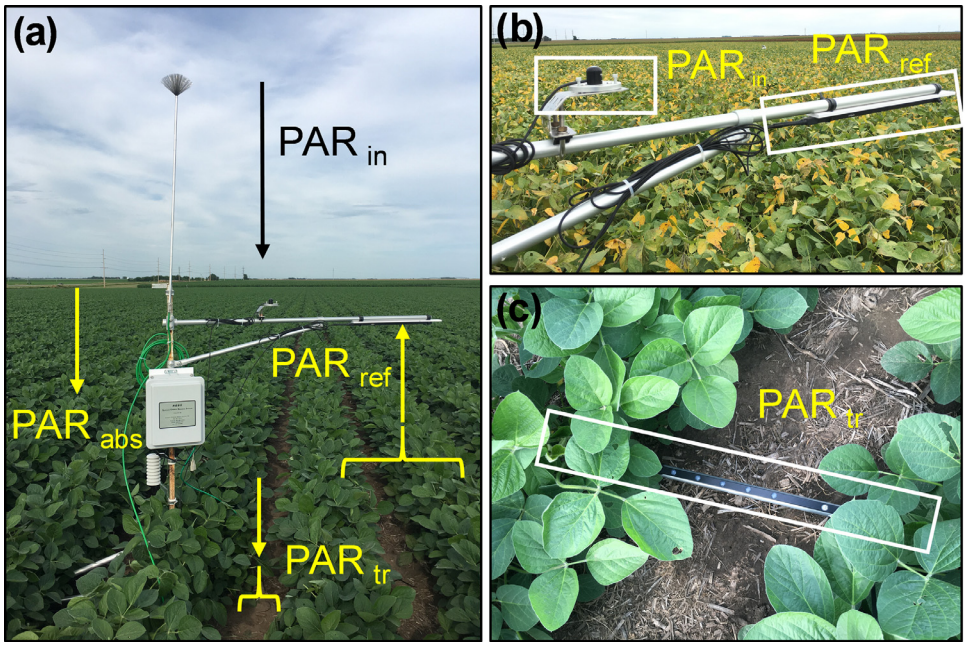


Fig. 2. The continuous Photosynthetically Active Radiation (PAR) balance systems monitoring light interactions mounted in soybean canopy. Identical setups were used to monitor the same variables in other crop canopies. The three constituent pictures show (a) various light flux components and their directions over any vegetative surface (soybean shown here); (b) point quantum flux and inverted line quantum flux sensor mounted over the soybean canopy to measure incoming PAR (PAR_{in}) (downwelling), and reflected PAR (PAR_{ref}) (upwelling), respectively; (c) A line quantum sensor at the soil surface level beneath the canopy to measure transmitted PAR (PAR_{tr}) (downwelling). All the sensors shown recorded light interactions every 15-min (15 1-min measurements were averaged).

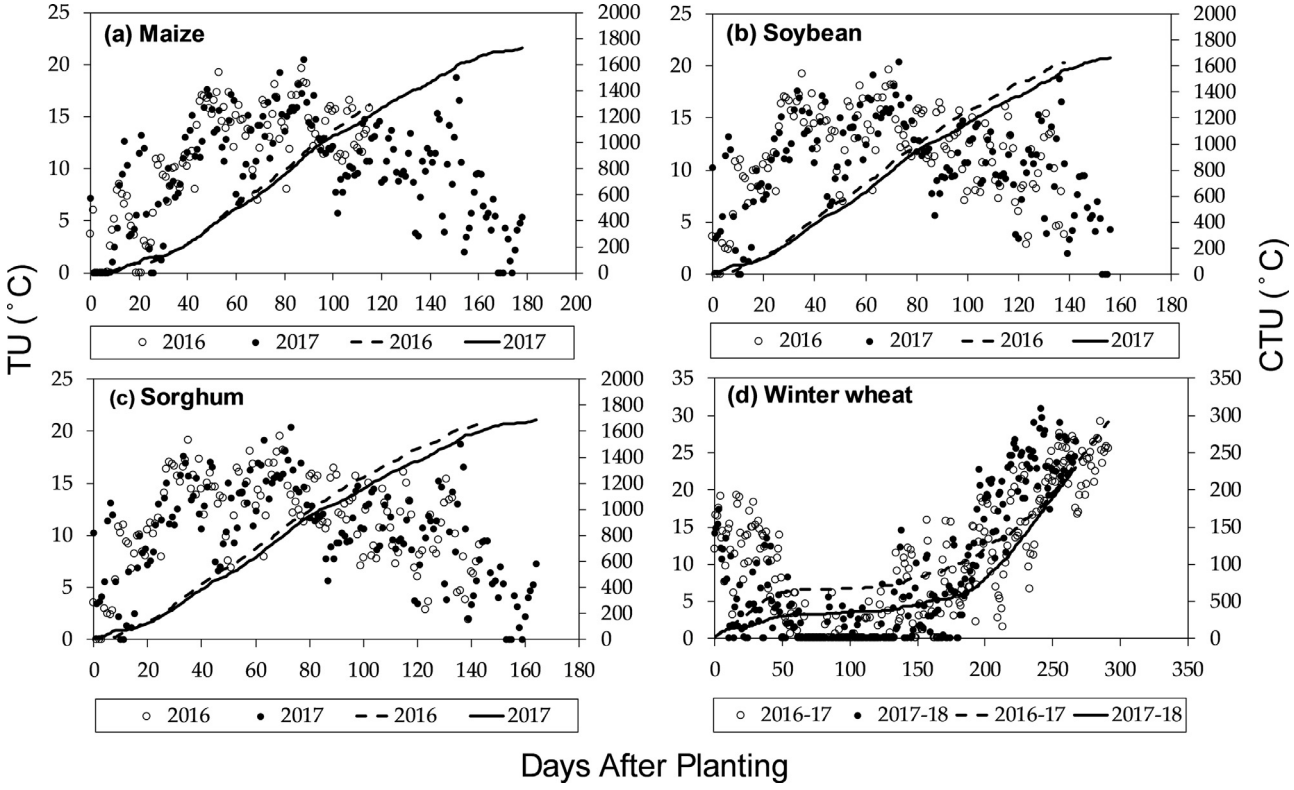


Fig. 3. Distribution of daily and cumulation thermal units (TU and CTU) observed during the two growing seasons in relation to days after planting for (a) maize; (b) soybean; (c) sorghum; and (d) winter wheat.

formulations for these indices and were used in this research:

$$T = \frac{PAR_{tr}}{PAR_{in}} \quad (2)$$

$$R = \frac{PAR_{ref}}{PAR_{in}} \quad (3)$$

Reflectance (R) can also be interpreted as a PAR-based albedo, similar to a traditionally-accepted albedo value, but uses PAR instead of global radiation. T and R were calculated on 15 min and daily time scales from the continuous light data and weekly (or longer) scale from ceptometer measurements.

Fractions of intercepted and absorbed PAR (fIPAR and fAPAR). fIPAR and fAPAR also serve to describe a vegetative surface for their light interaction features and are more widely used in the literature than T and R. fiPAR and fAPAR are used as important inputs in LUE models to quantitatively convey the phenological/developmental stage and condition. Eqs. (4) and (5) provide the formulations for these derived

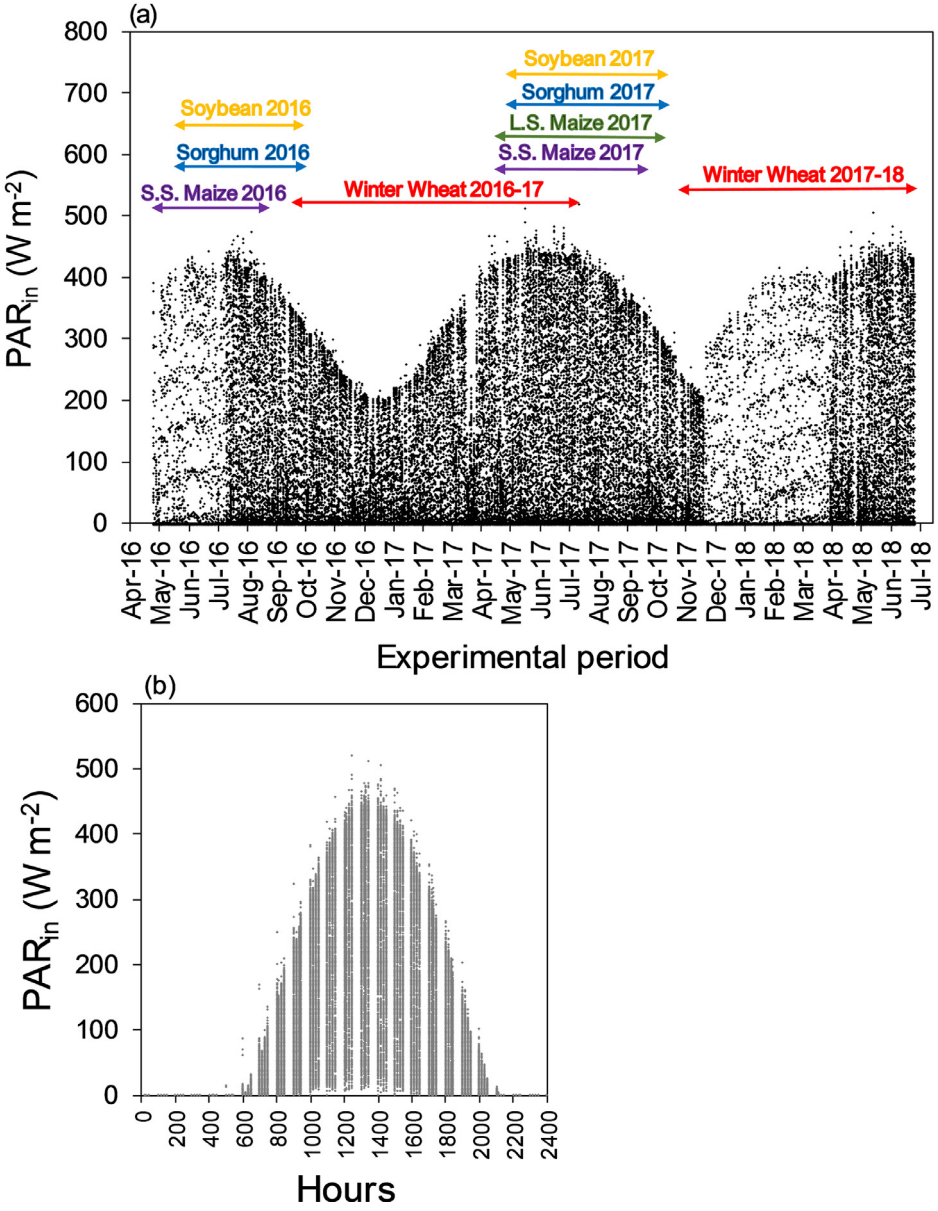


Fig. 4. (a) Sub-hourly PAR_{in} data measured at the research site over the complete duration of research investigation. The colored horizontal arrows above the data denote the temporal extent of the two growing seasons for each crop that were included in this research. (b) Diurnal patterns (15-min intervals throughout the day) of sub-hourly PAR_{in} measured at the research site for the complete research duration.

indices. fIPAR and fAPAR were calculated on 15-min and daily time scales from the continuous light data and weekly (or longer) time scale from ceptometer measurements:

$$fIPAR = \frac{PAR_{in} - PAR_{tr}}{PAR_{in}} \quad (4)$$

$$fAPAR = \frac{PAR_{in} - (PAR_{tr} + PAR_{ref})}{PAR_{in}} \quad (5)$$

2.3.4.3. Canopy light extinction coefficient. Sub-seasonal canopy light extinction coefficients (k) were calculated from T and LAI as described in Eq. (6). These were calculated for the days in the growing season when LAI and T were measured concurrently using a ceptometer. To estimate seasonal k, a linear regression of natural log (ln) of (1-fIPAR) and LAI was conducted, and the slope of this linear relationship was interpreted as k:

$$k = -\ln \frac{T}{LAI} \quad (6)$$

2.3.5. Light use efficiency

LUE was individually estimated on the basis of both intercepted and absorbed PAR. Total aboveground biomass (AGB) accumulated starting from the first sampling date was regressed against the cumulative (a) intercepted PAR and (b) absorbed PAR during the same period. The slope derived from (a) was LUE based on intercepted PAR (LUE_i) and that from (b) was LUE based on absorbed PAR (LUE_a).

3. Results and discussion

3.1. Weather conditions at the research site

The two experimental growing seasons were assessed for their similarities and dissimilarities of weather conditions and for their potential role in governing differences in plant growth and development

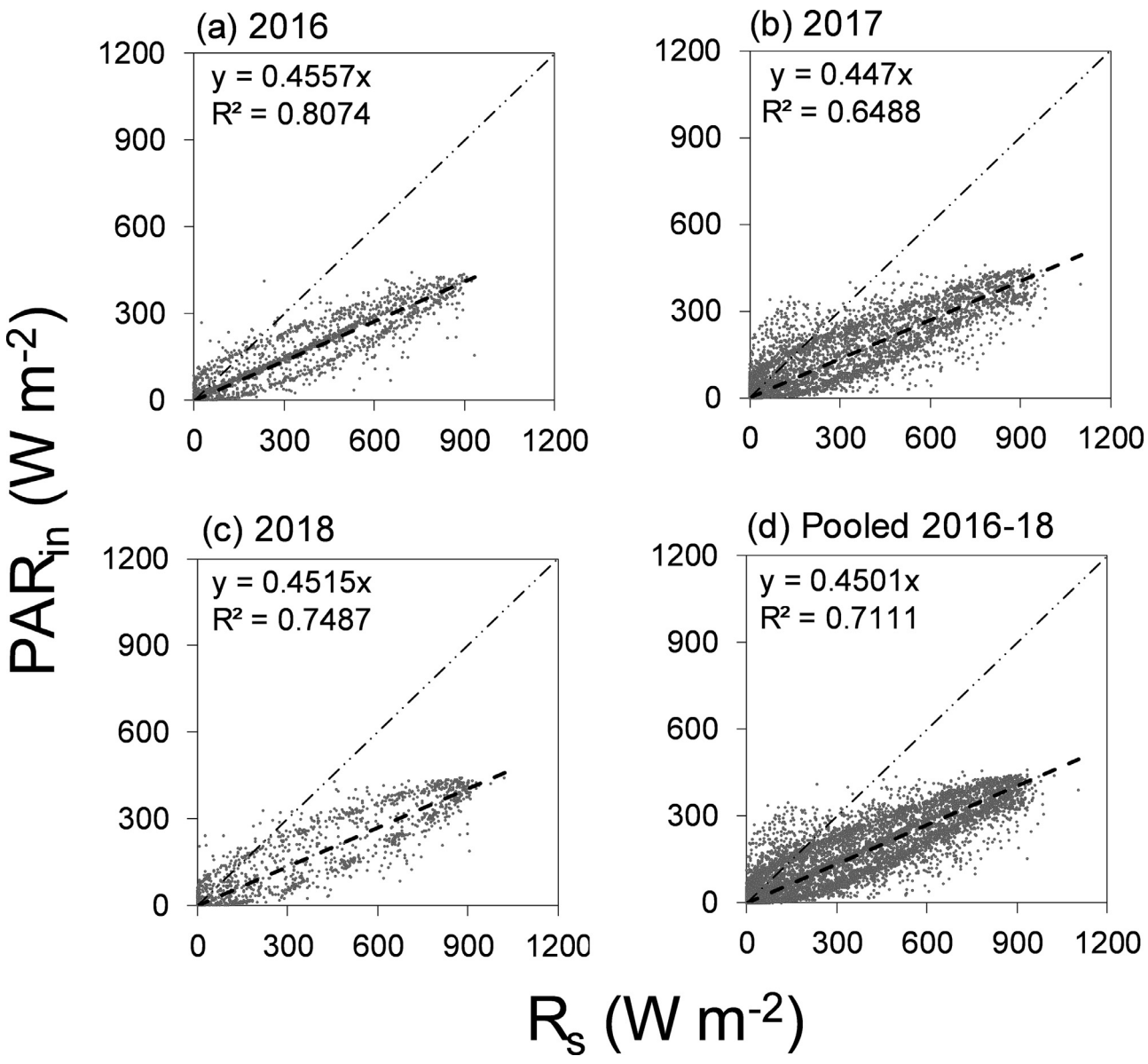


Fig. 5. Regression analysis of hourly PAR_{in} and R_s (incoming global radiation) received at the research site using (a) 2016 data; (b) 2017 data; (c) 2018 data; and (d) pooled data for the research duration. The intercept was forced to origin since it was not significantly significant from zero. The slope (darker broken line) represents the mean portion of R_s that is received as PAR_{in}. The lighter alternating pattern broken line represents a 1:1 fit.

across the two cropping cycles, if there were any. In this regard, the most crucial variable is heat accumulation, represented by cumulative thermal units computed for both growing seasons of the four crops, since the day of planting, and are presented in Fig. 3. Heat available for growth and development for each crop since planting (and emergence) was computed at daily time scale and was represented by cumulative growing degree days [referred by cumulative thermal units (CTU) in this research]. In this research, CTU values were calculated using the most widely accepted method outlined by McMaster and Wilhelm (1997). Base temperatures adopted were 10 °C for maize, soybean, and sorghum, and 0 °C for winter wheat. It was observed that heat accumulation was similar across two growing seasons for the three summer crops (maize, soybean and sorghum). This is in contrast to winter wheat, for which the CTU for the 2017–2018 growing seasons were substantially lower than 2016–2017 season since early in the fall and these differences existed until 200 DAP. Following 200 DAP, the air temperatures in spring 2018 were unusually high, driving CTU to magnitudes closer to those in 2016-2017 season. However, crop growth was particularly fast and hence harvested at lower total CTU than in 2016-2017 season. The total CTU values accumulated during the complete crop season (since planting and emergence) are listed in Table 1. The research included a short as well as a long season maize hybrid in 2017, which differed in timing to physiological maturity. As a result, the total CTU at harvest for the L.S. maize were 150 °C higher than that for S.S. maize. Total CTU in S.S. maize was also higher for 2017 than 2016, but this was only a result of a later harvest due to slightly lower TU per day (0.75 °C) in 2017, resulting in more days required to attain optimum grain moisture content for harvest.

Precipitation received during the growing season was greater in 2017 than 2016 for S.S. maize (by 23%), sorghum and soybean (by 37%), whereas 2016-2017 growing season was wetter than 2017-2018 for winter wheat (by 27%). Nevertheless, these precipitation amounts were supplemented by irrigation (amounts listed for each growing season in Table 1), and hence, the crops did not experience any water stress. This was ensured by monitoring the soil moisture throughout the root profile in all crops at multiple locations, and available soil water was never allowed to deplete beyond the maximum allowable depletion (40-45% of total available water). For detailed information on water

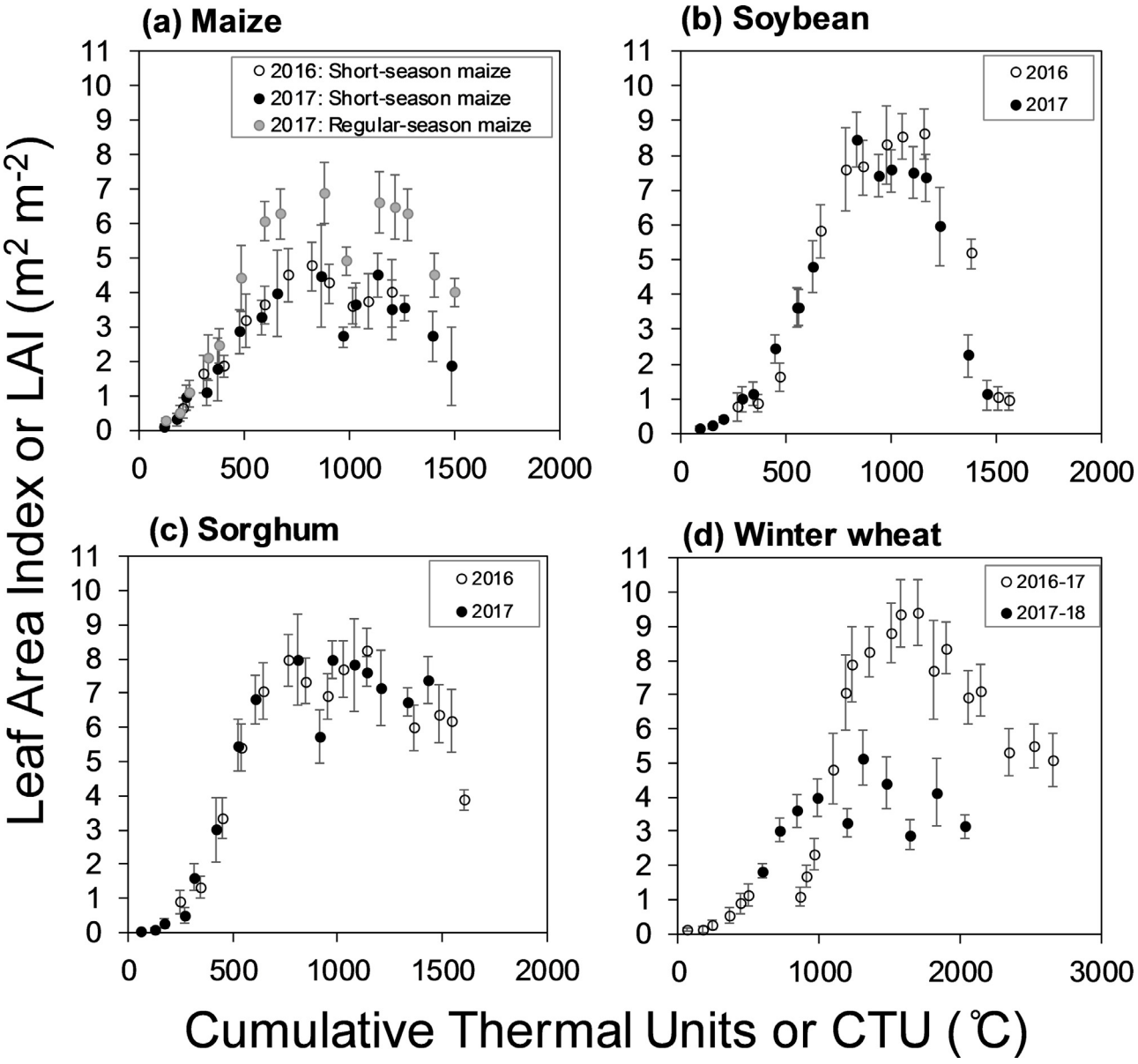


Fig. 6. Seasonal progression of leaf area index (LAI) during the two growing seasons in relation to cumulative thermal units (CTU) for (a) maize; (b) soybean; (c) sorghum; and (d) winter wheat. The error bars represent the standard deviation of 30 measurements conducted spatially in each crop.

use dynamics of these crops, refer to Kukal and Irmak (2020).

3.2. Incoming light (PARin) at the research site

The distribution of sub-daily PAR_{in} during this period at the research site is presented in Fig. 4(a), with horizontal lines marking the temporal extent of crop-specific growing seasons. The annual extremes of maximum PAR_{in} were around 450 W m^{-2} and 200 W m^{-2}, which occurred in July and December, respectively. PAR_{in} was also subjected to diurnal trends, with the peak occurring around 12:15-12:30 h (implies solar noon), with values declining gradually on either side of solar noon, and eventually gets to 0 at sunrise and sunset [Fig. 4(b)]. The highest values in the curve at each hour represent a typical sunny (or cloud cover) day in July, and the distribution changes if cloudy/overcast conditions exist or seasonal variation in daylength occurs. The PAR_{in} was also studied in comparison to incoming global radiation (R_s) (presented in Fig. 5) and it was found that the slope of the regression between the two when intercept was set to zero (since the intercept approached zero) was 0.45 for each individual year, as well as when all the data were pooled. Hence, it was established that PAR_{in} was 45% of the R_s at the experimental site, and this relationship was used to fill two distinct gaps in the PAR_{in} dataset from R_s (May to early July 16, prior to installation of PAR balance systems, and another one in the dormant season of winter wheat in the 2017-2018 growing season). The lower datapoint densities in these time spans (Fig. 4(a)) are due to the hourly (not 15 min) resolution of the R_s data used to fill in PAR_{in} gaps.

3.3. Leaf area index and aboveground biomass

Each crop showed a distinct pattern of seasonal progression of LAI magnitudes and the maximum LAI value and the timing of maximum LAI (Fig. 6). The maximum value of LAI in the growing season was shown by winter wheat, followed by soybean, sorghum, L.S. maize and S.S. maize (Table 2). LAI values for the two growing seasons were similar in magnitudes in each crop, except winter wheat, which in 2017–2018 growing season demonstrated around 50% reduced LAI, owing to late planting due to high fall precipitation and the abrupt increase in CTU around mid-growing season. LAI reached its maximum

Table 2
Descriptive statistics of crop growth and light interactions and use efficiency for each crop.

Variable	Units	Crop				
		S.S. maize	L.S. maize	Soybean	Sorghum	Winter wheat
AGB at harvest	g m^{-2}	3503	4713	1946	3065	2762
Peak LAI	m^2 m^{-2}	4.76	6.88	8.62	8.25	9.41
Mean T	fraction	0.34	0.29	0.27	0.12	0.51
Minimum T	fraction	0.08	0.03	0.01	0.00	0.03
Mean R	fraction	0.06	0.04	0.05	0.05	0.06
Minimum R	fraction	0.03	0.02	0.03	0.03	0.03
Maximum R	fraction	0.12	0.12	0.11	0.15	0.11
Mean fIPAR	fraction	0.66	0.71	0.73	0.88	0.50
Maximum fIPAR	fraction	0.92	0.97	0.99	1.00	0.97
Mean fAPAR	fraction	0.60	0.66	0.68	0.82	0.46
Maximum fAPAR	fraction	0.88	0.95	0.96	0.96	0.95
DAE until 80% fAPAR	days	56	49	56	40	190
CTU until 80% fAPAR	°C	618	490	694	497	874
LAI at 80% fAPAR	m^2 m^{-2}	3.4	4.4	5.7	4.6	3.7
DAE until 90% fAPAR	days	-	57	61	53	201
CTU until 90% fAPAR	°C	-	596	778	693	1027
LAI at 90% fAPAR	m^2 m^{-2}	-	6.1	6.7	6.3	5.6
k	dimensionless	0.45	0.46	0.48	0.48	0.36
CV of sub-seasonal k	Percent	14	15	18	18	35
Slope of sub-seasonal k	day^{-1}	0.0011	0.0012	0.0017	0.0014	0.0014
LUE_a	g MJ^{-1}	5.21	5.37	3.06	4.13	4.05
LUE_i	g MJ^{-1}	4.82	5.10	2.84	3.88	4.00
CV of sub-seasonal LUE_a	Percent	32	30	33	42	54

Abbreviations used: AGB (aboveground biomass); LAI (leaf area index); T (transmittance); R (reflectance); fIPAR: (fraction of intercepted photosynthetically active radiation); fAPAR: (fraction of absorbed photosynthetically active radiation); k (light extinction coefficient); LUE_a (light use efficiency based on absorbed light); LUE_i (light use efficiency based on intercepted light); CV (coefficient of variation).

value earliest (or at lowest CTU) for maize, followed by sorghum, soybean and winter wheat. The comparative magnitudes of LAI in different crops at various times during the season represent different levels of interception of incident light and hence the comparative levels/biomass of carbon assimilatory tissue in different crops. The spatial variability associated with LAI (shown by standard deviation bars in Fig. 6) was the lowest in winter wheat, followed by sorghum, soybean, L.S. maize and S.S. maize, when evaluated using a coefficient of variation (CV) metric (data not shown) and is an indicator of relative homogeneity in canopy structure. Also, the CV in LAI was observed to be higher in the initial part of the growing season (lesser homogenous canopy conditions), and gradually tapered to a lower value (more homogenous canopy conditions) later in the growing season for all crops (data not shown here).

Aboveground biomass (AGB), which was observed using destructive sampling, shows a pattern of linear growth until a plateau is reached when the crop reaches physiological maturity (Fig. 7). The greatest mean AGB at physiological maturity was obtained in L.S. maize, followed by S.S. maize, sorghum, winter wheat and soybean (Table 2). As was observed in the case of LAI, the two growing seasons were consistent in their biomass production for all crops, except winter wheat, which produced relatively lesser biomass in the 2017-2018 growing season. This lower biomass production is also reflected and explained by the lower (by one-half) winter wheat LAI that was observed in 2017-2018 growing season than in 2016-2017 season, although the biomass at physiological maturity was similar across the two growing seasons. The LAI and AGB values measured in this research are generally higher than those reported in the literature, especially for soybean, sorghum and winter wheat (only in 2017-2018), and to the best of our knowledge, these crops have not been shown to have LAI values of greater than 6-7 m^2 m^{-2}. In Nebraska, Lindquist et al. (2005) recorded maize LAI (green LAI measured via a planometer) that was comparable to that found in our research, but only for L.S. maize. However, the AGB of L.S. maize recorded in our research (both S.S. and L.S. maize) was considerably higher than that reported by Lindquist et al. (2005). The high productivity can be explained by the optimal sufficiency of crop inputs (water and nutrients) and growing conditions of the research settings, typical high productivity soil and climatic characteristics of south central Nebraska and the current hybrids and varieties. Kukal and Irmak (2019) provides greater quantitative details of crop growth in these experiments.

3.4. Light transmittance

Light transmittance (T) was calculated using the continuous PAR balance data at two time scales: 15 min and daily. While the 15 min T represents the sub-daily variability encountered in T, the daily T was calculated using the daily sums of PAR_{tr} and PAR_{in}, from 15 min data. Additionally, the manual (ceptometer-based) light measurements were also used to calculate T at roughly on a weekly interval. All three scales of T measurements are presented for the two entire growing seasons in Fig. 8 for each crop. It should be noted that in some cases, the manual light measurements were the only source of T data, for example in L.S. maize in 2017, and initial part of the S.S. maize in 2016, soybean in 2016 and 2017 and sorghum in 2016. Nevertheless, both sources (manual and continuous) were combined to form a complete seasonal T profile for each crop. Generally across all crops, T is the highest (approaching 1 theoretically just prior to emergence) in the early growing season, and thereby decreases as the canopy develops, followed by a plateau for the duration of full canopy cover, and starts to finally increase to a magnitude that is crop-dependent as canopy senesces. Despite this general trend, various crops have distinguishable features in their seasonal T profiles. For example, the seasonal T profile in soybean follows an inverted bell-shaped curve, while that in sorghum stays linear since full cover until physiological maturity (PM), and maize and winter wheat demonstrate a 10-20% increase in T at senescence. These differences are a function of varying physiological attributes of these crops, wherein soybean canopy completely and quickly defoliates at physiological maturity (causing higher *T* values), while sorghum canopy transmits minimal light until PM owing to the presence of the "stay green" trait (Walulu et al., 1994), and finally, maize and winter wheat demonstrate partial defoliation and leaf senescence. Winter wheat is also distinct in its typical two-peak T profile (observed in 2016–2017 growing season), owing to pre-dormancy and post-

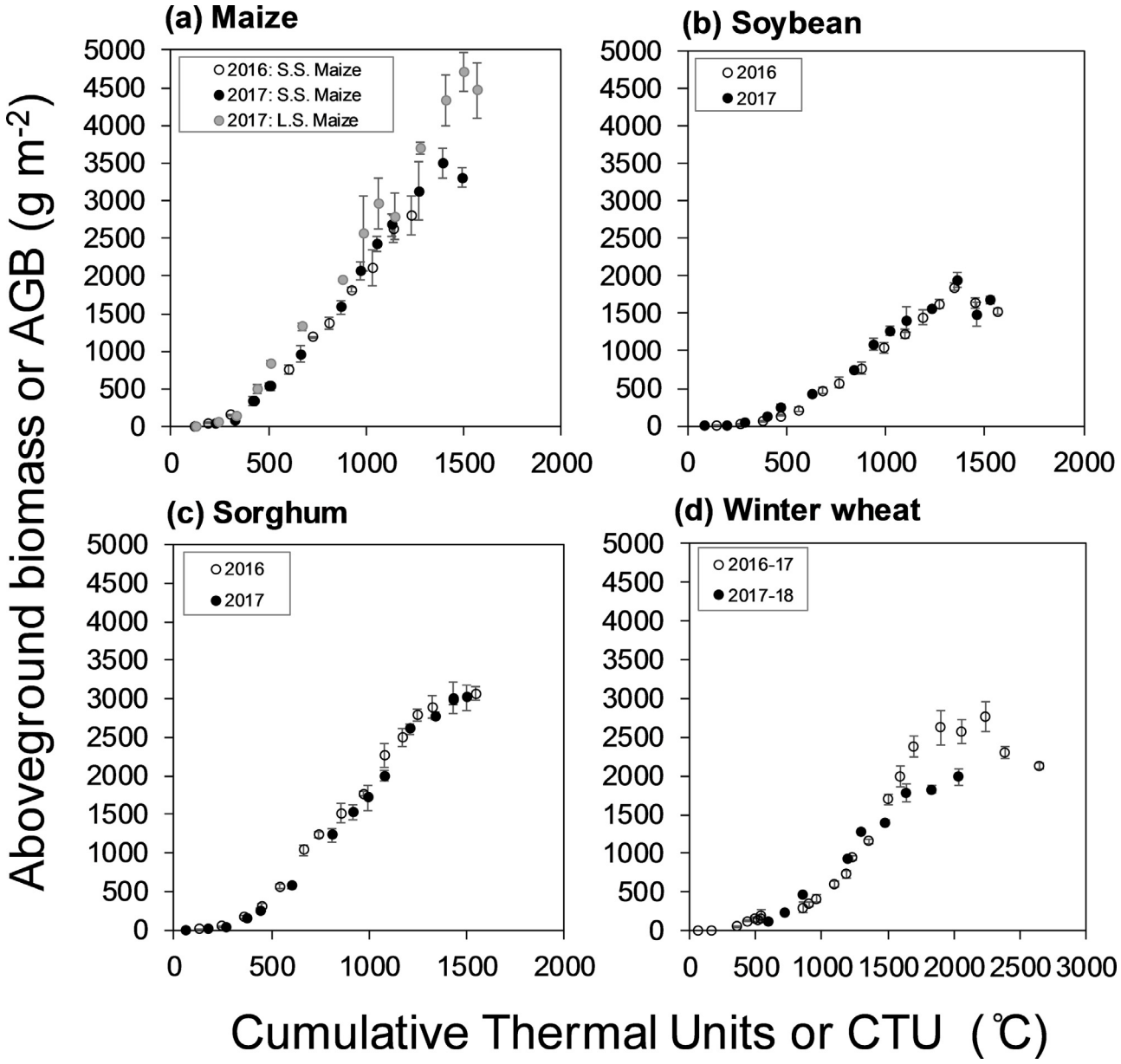


Fig. 7. Seasonal progression of aboveground biomass (AGB) during the two growing seasons in relation to cumulative thermal units (CTU) for (a) maize; (b) soybean; (c) sorghum; and (d) winter wheat. The error bars represent the standard deviation of 30 measurements conducted spatially in each crop.

dormancy growth periods, although pre-dormancy growth is considerably lower and is subject to timing of frost occurrence (2017-2018 winter wheat did not have significant pre-dormancy growth owing to lower temperatures and earlier first fall frosts). Comparing across crops, soybean, sorghum, winter wheat and L.S. maize had lowest and similar *T* values during full canopy cover, although for different periods of time, while S.S. maize had significantly higher *T* values. Table 2 lists mean and minimum T values measured during the growing season for each crop, and the mean T is function of the complex interactions of the aforementioned crop physiological attributes. For example, although winter wheat showed lower minimum *T* than S.S. maize, the mean *T* was lower in S.S. maize due to differences in the period of low *T* within the growing season (Fig. 8).

3.5. Light Reflectance

Light Reflectance (R) was computed and presented in the same manner as *T* in Fig. 9. Unlike *T*, which can assume a complete range of possible values, i.e., 0 to (approaching) 1, it was found that PAR_{ref} was always $<20\%$ relative to PAR_{in} ($R<0.2$), on a daily basis (red circles in Fig. 9) and was subject to little variation as the canopy size and structure changed. A general trend in R was revealed and is characterized by higher values of R early and late in the season, and minimal values were observed at full canopy cover. The seasonal R profile follows a trough-like curve, with the exception of winter wheat in 2016-2017, where two troughs are observed, each corresponding to a pre-and post-dormancy growth period. The extremes of R across crops are mostly uniform, and both the magnitude and seasonal profiles are non-distinct to various crops. This is most likely due to largely similar magnitude of reflection of PAR_{in} occurring from the canopy, and rather, R must be a function of exposed soil surface. This confirms the higher R values observed in the early growing season during partial canopy cover and during senescence when soil surface is also exposed. Thus, R is dependent on the crop-specific timing and rates of canopy closure and senescence (which eventually governs exposed surface), instead of the canopy growth itself. This is further confirmed by the absence of day-to-

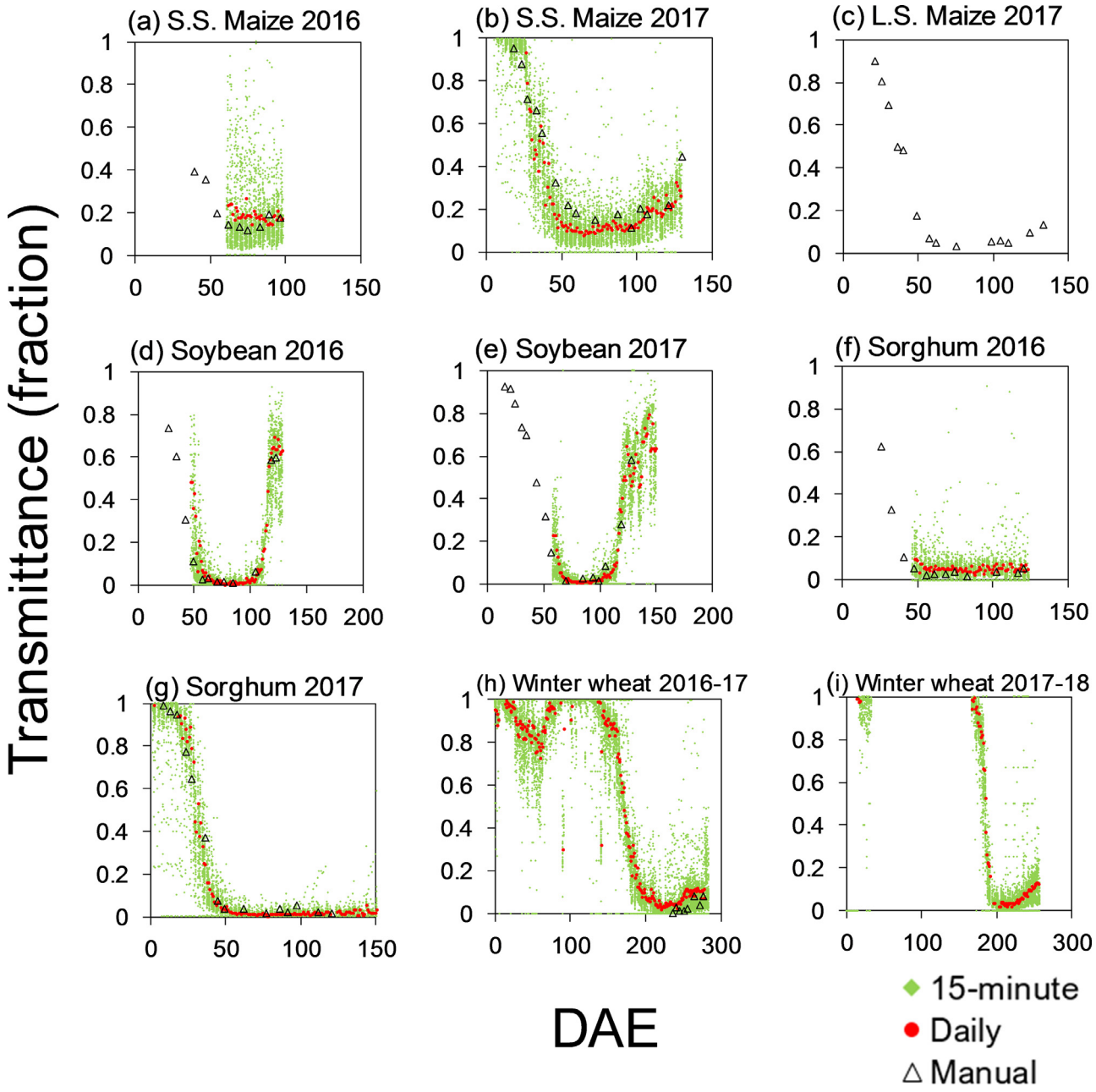


Fig. 8. Seasonal profiles of transmittance (T) for both growing seasons of (a) S.S. maize 2016; (b) S.S. maize 2017; (c) L.S. maize 2017; (d) soybean 2016; (e) soybean 2017; (f) sorghum 2016; (g) sorghum 2017; (h) winter wheat 2016-17; (i) winter wheat 2017-18. Three different series are used to represent the seasonal profiles, i.e., 15-min average data (green diamonds), 15 min data aggregated to daily scale (red circles), and manual (instantaneous) ceptometer measured data (hollow triangles) (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

day fluctuations in R that were observed in the seasonal profile of T. The extremely high daily R values observed in winter wheat arise from the snow cover, which reflects a substantial portion of PAR_{in}, and must not be confused by R due to the vegetative surface. The crop-specific mean, maximum and minimum values of R observed during the crop growing season are listed in Table 2. Overall, it was observed that R is a relatively small portion of the PAR balance, when compared to T.

3.6. Light interception and absorption

The fraction of PAR intercepted and absorbed by the crop canopies are represented using fIPAR and fAPAR, respectively, and are shown in similar manner as previous indices in Figs. 10 and 11, respectively. The fIPAR curve is derived from that of T, and hence follows the same seasonal dynamics as T, but in an inverse fashion. This is simply because interception is the portion of PAR_{in} that did not reach the ground surface, and hence, a high T implies low interception, and vice versa. However, fAPAR curve integrates both T and R curves into a single variable and reports the actual proportion of PAR_{in} that gets absorbed by the canopy. Consequently, fAPAR, at any time in the growing season, is always lower than fIPAR. The fIPAR and fAPAR seasonal profiles are mostly similar despite the difference in magnitude, although days in the early and late growing season are exceptions. This is because of accounting for R in fAPAR, unlike fIPAR, and it has been shown earlier that R has relatively large values in these early and late stages of the growing season. fIPAR, due to its consideration of light transmission, is conceptually similar to LAI, and hence has the potential to, and has been used to convey ground cover quantitatively. The parallel

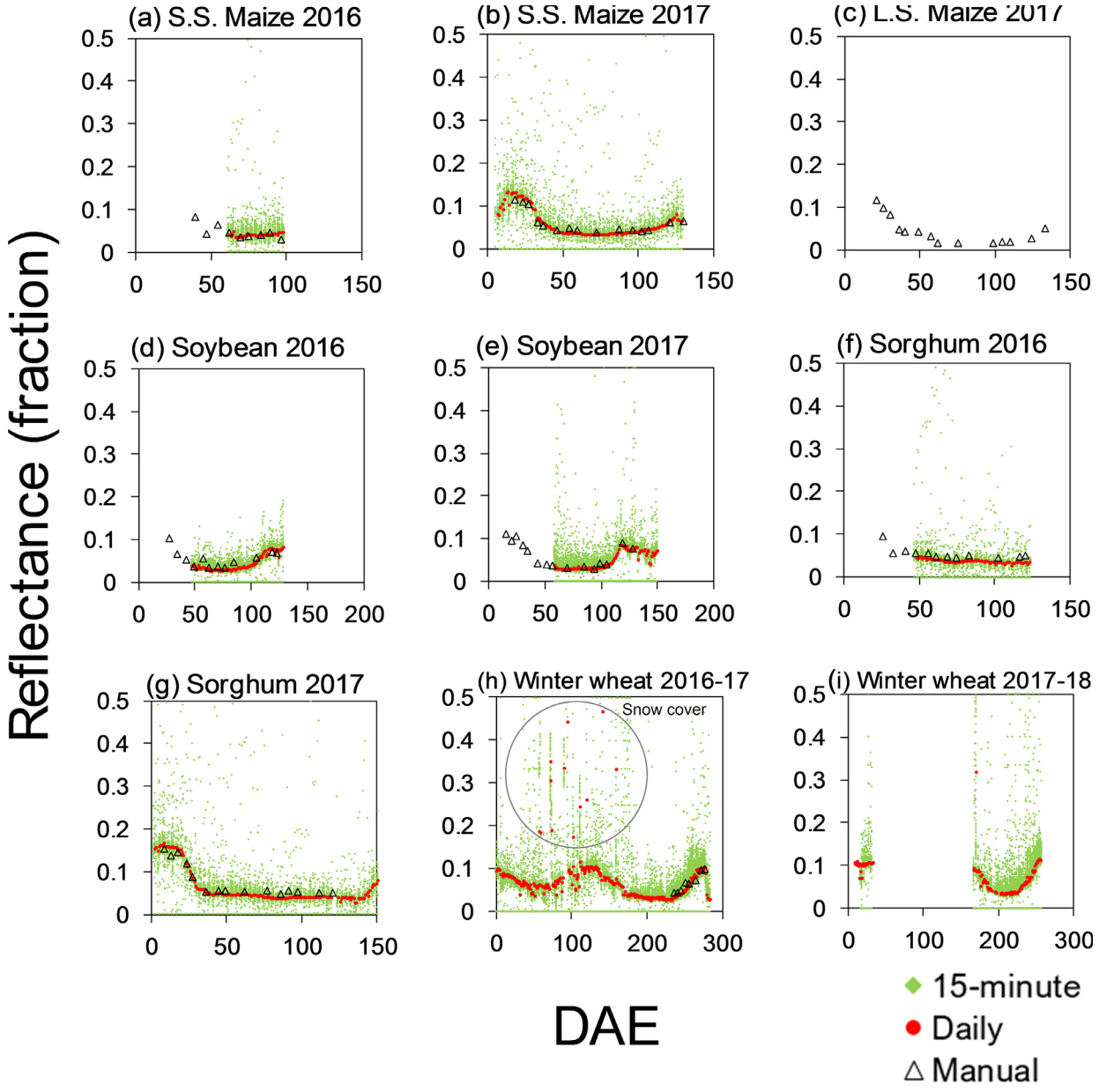


Fig. 9. Seasonal profiles of reflectance (R) for both growing seasons of (a) S.S. maize 2016; (b) S.S. maize 2017; (c) L.S. maize 2017; (d) soybean 2016; (e) soybean 2017; (f) sorghum 2016; (g) sorghum 2017; (h) winter wheat 2016–17; (i) winter wheat 2017–18. Three different series are used to represent the seasonal profiles, i.e., 15 min average data (green diamonds), 15 min data aggregated to daily scale (red circles), and manual (instantaneous) ceptometer measured data (hollow triangles) (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

measurements of fIPAR and LAI conducted in this research provide a unique opportunity to develop crop-specific empirical relationships between these two indices (Fig. 12). The relationship was linear until a point around LAI of 2-4, and then transitioned to a diminishing return curve. This signifies that addition of morphological plant matter (leaf area) increases light interception linearly until a threshold, and the light interception increments diminish by further addition of morphologic components, especially after the canopy is intercepting 85–90% of PAR_{in}. For future use, we also propose and recommend the use of the crop-specific fIPAR vs. LAI relationships in Table 3. The specific characteristics of crop-specific light interception (absorption) were quantified using some metrics such as maximum and mean fIPAR and fAPAR observed during the growing season, and timing (DAE), heat accumulation required, and LAI to reach 80 and 90% rates of light absorption (fAPAR = 80 and 90%, respectively) and are listed in Table 2. Sorghum showed the highest mean fIPAR and fAPAR, followed by soybean, L.S. maize, S.S. maize and winter wheat, and each crop intercepted and absorbed at least 92 and 88% of the PAR_{in}, at their peak interception and absorption duration, respectively, (Table 2). Also, it was revealed from a quantitative analysis of the fAPAR seasonal profiles, that sorghum reached 80% fAPAR earliest, followed by L.S. maize, S.S. maize, soybean and winter wheat (in terms of DAE), while L.S. maize required the least CTU to reach 80% fAPAR followed by sorghum, S.S. maize, soybean and winter wheat. It should be noted that while the three summer crops required <60 days to reach 80% fAPAR, winter wheat required 190 days, which is attributed to inclusion of dormant season in this metric, and also the fact that most of the winter wheat growing season is characterized by lower thermal units accumulated per day, relative to summer, and hence, CTU is a better metric to compare equivalent time to reach a certain light absorption across crops. Moreover, it was found that the LAI at the time of 80% fAPAR (Table 2) for each crop demonstrated a wide range (difference of 2.2. m^2 m^{-2},

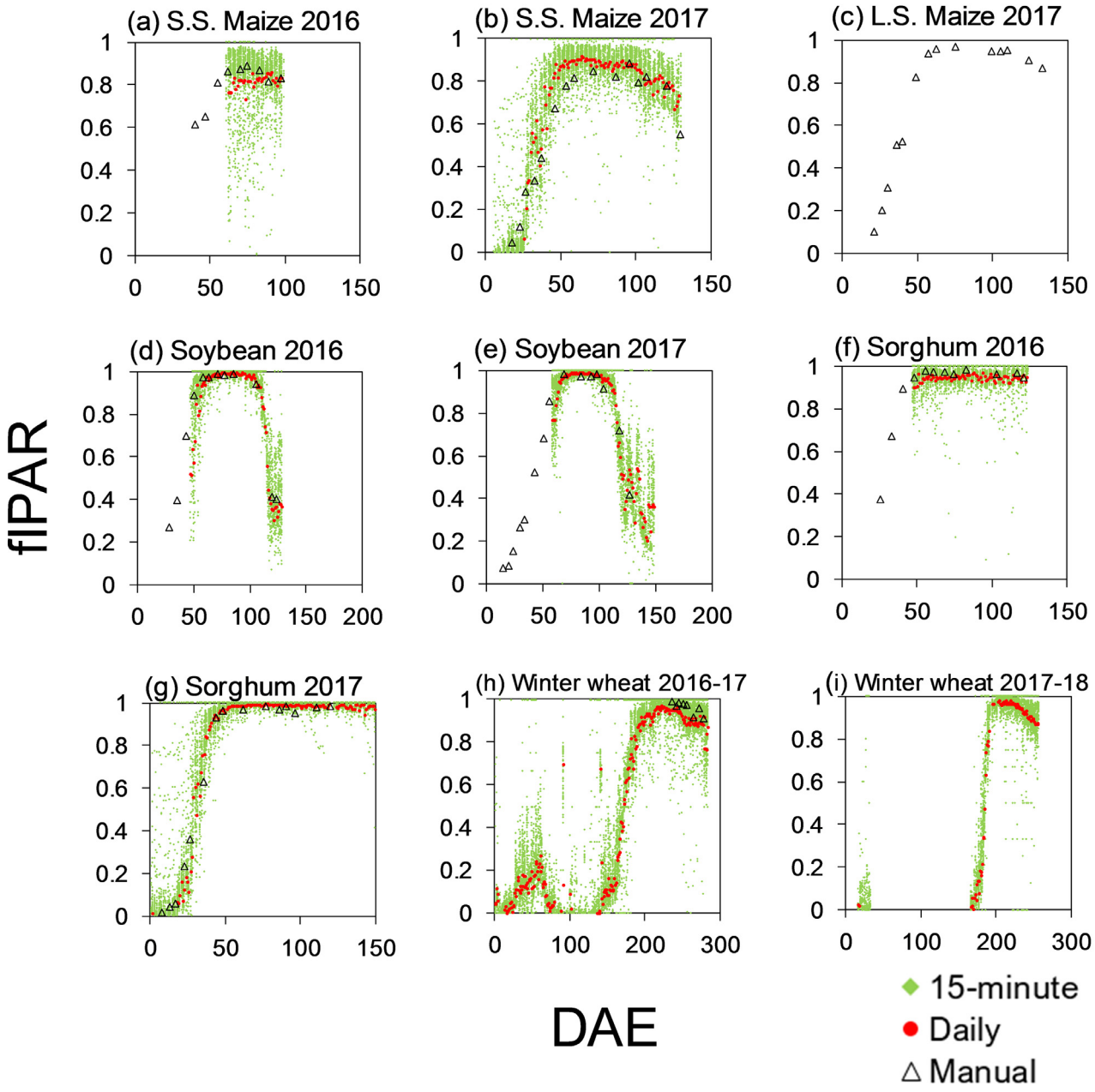


Fig. 10. Seasonal profiles of fraction of intercepted photosynthetically active radiation (fIPAR) for both growing seasons of (a) S.S. maize 2016; (b) S.S. maize 2017; (c) L.S. maize 2017; (d) soybean 2016; (e) soybean 2017; (f) sorghum 2016; (g) sorghum 2017; (h) winter wheat 2016-17; (i) winter wheat 2017–18. Three different series are used to represent the seasonal profiles, i.e., 15 min average data (green diamonds), 15 min data aggregated to daily scale (red circles), and manual (instantaneous) ceptometer measured data (hollow triangles) (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

with the lowest and highest LAI shown by S.S. maize (3.4 m^2 m^{-2}) and soybean (5.7 m^2 m^{-2}). Similarly, comparison of days, TU and LAI required to reach 90% fAPAR had crop-specific magnitudes and the metrics are listed in Table 2. It has to be noted that in this research, we are assuming CTU to the sole driver of crop development, and light use, and neglecting other minor impacts such as vernalization conditions for winter wheat, wind effects etc. Moreover, our considerations of crop-specific base temperatures are generic commonly used values and not cultivar-specific, which is a limitation, but is also a challenging task to accomplish in field conditions.

Both fIPAR and fAPAR are utilized to denote canopy cover and canopy light interactions, however, not all research quantify both indices, mostly due to failure to measure canopy reflection (especially when manual measurements are conducted). This induces the need for a framework for cross-conversion from fIPAR and fAPAR and vice-versa, in order to compare light dynamics among different research findings. We calculated the ratio of fAPAR and fAPAR measured daily for each crop and revealed the non-linear nature of this ratio when studied against LAI during the growing season (Fig. 13). The ratio was around 95-98% for all crops (98% for L.S. maize, 97% for soybean and winter wheat, 96% for S.S. maize and sorghum) for the most part of the growing season, especially when the LAI was higher than a certain threshold. This threshold, which was distinct for all crops, occurred earliest in winter wheat (LAI of 1.4), maize (LAI of 2.8), sorghum (LAI of 4.0) and soybean (LAI of 5). Preceding this threshold value, the ratio was substantially lower than what was observed following the

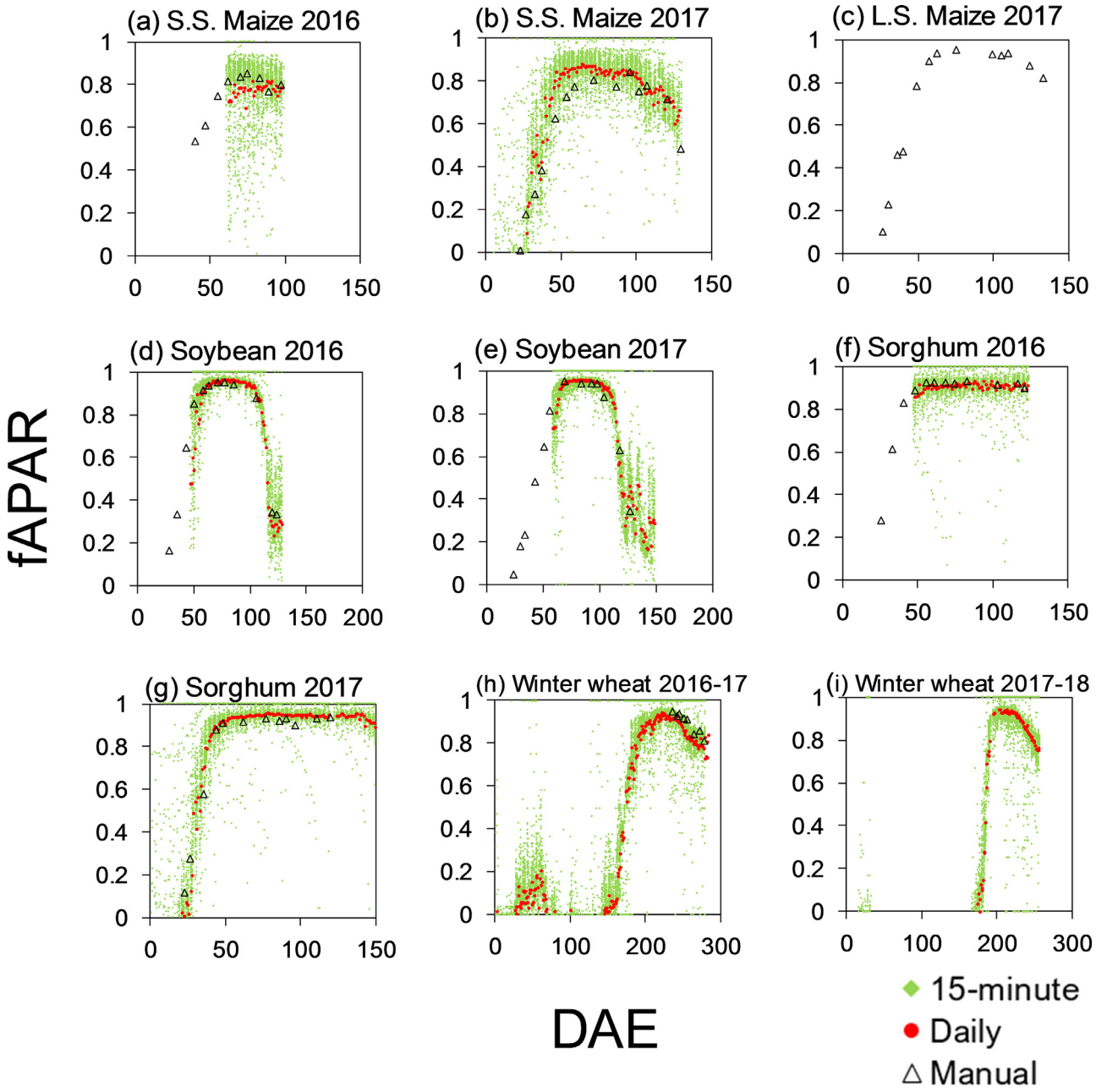


Fig. 11. Seasonal profiles of fraction of absorbed photosynthetically active radiation (fAPAR) for both growing seasons of (a) S.S. maize 2016; (b) S.S. maize 2017; (c) L.S. maize 2017; (d) soybean 2016; (e) soybean 2017; (f) sorghum 2016; (g) sorghum 2017; (h) winter wheat 2016-17; (i) winter wheat 2017-18. Three different series are used to represent the seasonal profiles, i.e., 15 min average data (green diamonds), 15 min data aggregated to daily scale (red circles), and manual (instantaneous) ceptometer measured data (hollow triangles) (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

threshold and is also highly sensitive to increase in LAI. In the past, Bonhomme (2000) and Sinclair and Muchow (1999) have shown that when LAI is large, it is accurate to assume that 85% of intercepted PAR is absorbed by the canopy, but the value is lower when canopy is less dense. We have shown that the usual assumption of the value of the ratio of fAPAR to fIPAR as 0.85, as suggested by these studies, are under-estimates by at least 12%, and for the row crops under consideration, under the given agronomic management practices at the experimental site. This LAI-dependent relationship among fIPAR and fAPAR also establishes that conversion of one to the other using a multiplicative factor may not be possible, and the best approach in this regard is to measure both intercepted and absorbed radiation continuously during the biomass sampling period, and report canopy light dynamics using both fIPAR and fAPAR. This would allow for greater transferability among canopy light dynamics based on either fIPAR or fAPAR and accurate comparisons with data in the literature. However, retrospectively, crop-specific relationships among the ratio of fAPAR to fIPAR vs. LAI (provided in Table 3) can be used for carrying out conversions, if information on either of the indicators and LAI is known.

3.6. The case for high-frequency light balance measurements

This research utilized both instantaneous light balance measurements (manual ceptometer-based), and 15 min continuous light balance measurements to sample dynamics, trends and patterns of canopy light interactions. Besides providing means for sampling the complete

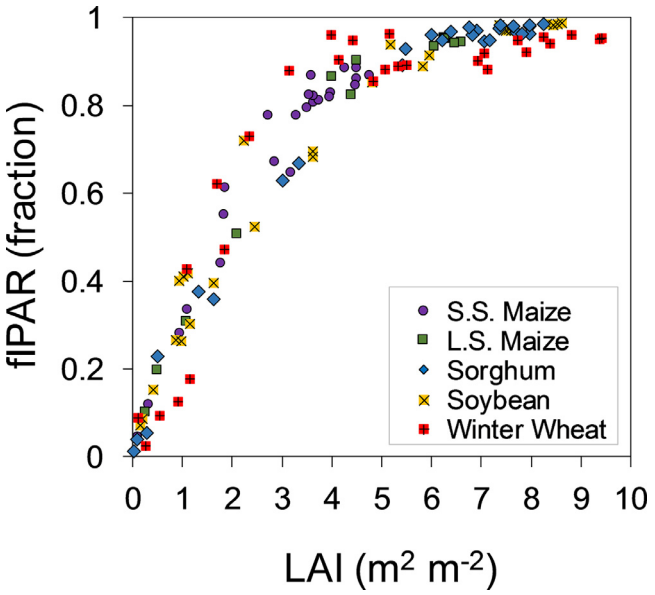


Fig. 12. Relationship between fraction of intercepted photosynthetically active radiation (fIPAR) and leaf area index (LAI) for each crop. The two measurements were conducted on clear-sky days using a ceptometer.

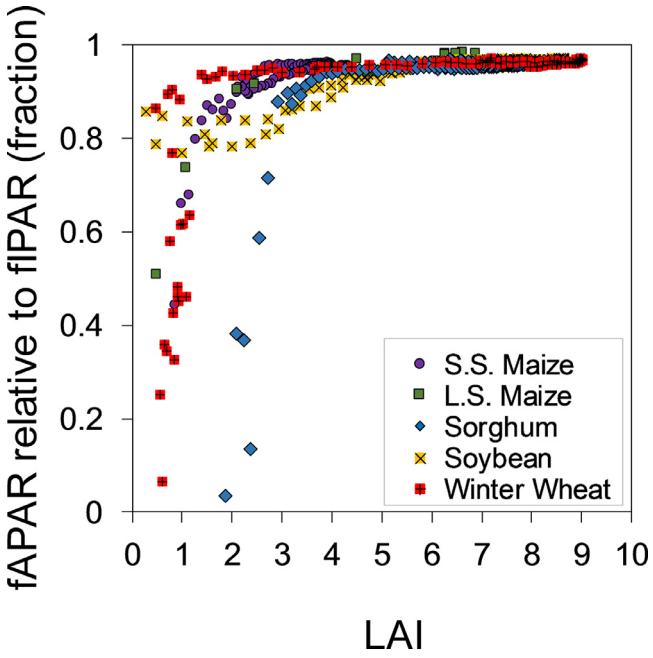


Fig. 13. Ratio of fraction of absorbed photosynthetically active radiation (fAPAR) to fraction of intercepted photosynthetically active radiation (fIPAR) in relation to leaf area index (LAI) for each crop. fIPAR and fAPAR were measured daily using continuous light balance data, while the LAI was interpolated using thermal units accumulated between any two sampling periods that were 1–1.5 weeks apart.

growing season and filling data gaps, the purpose of this was to determine how well does the instantaneous sampling approach replicate the continuous sampling approach, or what are the limits of the former relative to the latter. To evaluate this, we regressed the instantaneous measurements of T, R, fIPAR and fAPAR, which were taken around local solar noon, against the same indices, but derived using continuous sampling. The indices from continuous sampling were derived by aggregating the 96 individual measurements (24 hours × 4 15 min intervals) conducted daily throughout the growing season, whereas the instantaneously measured indices represent a short time window (<1 min for a single measurement) around solar noon, when ceptometer-based sampling was conducted in a day. The relationships presented in Fig. 14 were developed by pooling measurements from all four crops, as the choice of crop was not a determinant of how accurately the two approaches estimated these indices. We found that although the indices derived from the approaches were highly correlated ($R^2 > 0.88$), the deviation of the instantaneously measured indices can be significantly different from the true index value (represented by continuous measurements). For example, the instantaneous fAPAR was subjected to a root mean squared error (RMSE) of 0.06 units, on an average, and these errors could be as high as 0.19 units (40%) at certain times during the growing season. Specifically, higher errors were found in the early and late in the growing season (for crops that demonstrated a decline in LAI). Since, the underlying reason for the deviation is low LAI, crops that show relatively lower LAI than others (like maize) are subject to higher deviations than others. This is evident from the higher deviation of the data trendline from the 1:1 line when T and R are low, and when fIPAR and fAPAR are high (Fig. 14).

These weaknesses observed in the instantaneous light balance sampling approach could also potentially arise from either the approach itself (i.e., a single sampling around solar noon) or the differences in instrumentation. To determine the appropriate reasoning, the continuous 15 min continuous data were subset for the 12:15–12:30 h (solar noon), and were evaluated for their representativeness of the true indices. By doing this, we nullify the instrumentation differences, since both datasets were collected from the same instruments, and hence, only approach-based differences remain. Fig. 15 presents regression among the same indices in Fig. 14 derived from solar noon sampling and true values of indices. The analysis yields similar results as with the instantaneous data, in terms of slopes (0.93–0.96) and correlations (0.90–0.92) observed. We also observe that the solar noon measurements are reasonably accurate during full canopy cover, but are subject to greater uncertainties at early and mid-ranges of T, fIPAR and fAPAR

Table 3
Details of empirical relationships developed among fIPAR and ratio of fAPAR to fIPAR to LAI.

Variables Dependent (Y)	Independent (X)	R^2	Crop	Function Type	Function
fIPAR	LAI	0.90	S.S. maize	Logarithmic	$y = 0.2563\ln(x) + 0.4524$
		0.95	L.S. maize	Logarithmic	$y = 0.2927\ln(x) + 0.3911$
		0.97	Soybean	Polynomial (2nd order)	$y = -0.0152x^2 + 0.2322x + 0.0976$
		0.99	Sorghum	Polynomial (2nd order)	$y = -0.017x^2 + 0.2537x + 0.0289$
		0.89	Winter wheat	Logarithmic	$y = 0.2581\ln(x) + 0.4285$
		0.90	Pooled (all crops)	Logarithmic	$y = 0.2477\ln(x) + 0.4472$
fAPAR / fIPAR	LAI	0.94	S.S. maize	Polynomial (4th order)	$y = -0.019x^4 + 0.2329x^3 - 1.0515x^2 + 2.1006x - 0.6501$
		1.00	L.S. maize	Polynomial (5th order)	$y = 0.0004x^5 - 0.0092x^4 + 0.0846x^3 - 0.381x^2 + 0.8532x + 0.1679$
		0.92	Soybean	Polynomial (3rd order)	$y = -0.0009x^3 + 0.0096x^2 - 0.0004x + 0.8$
		0.93	Sorghum	Polynomial (5th order)	$y = 0.0006x^5 - 0.0193x^4 + 0.2515x^3 - 1.6047x^2 + 5.0146x - 5.1708$
		0.69	Winter wheat	Polynomial (5th order)	$y = 0.0001x^5 - 0.0046x^4 + 0.0562x^3 - 0.3282x^2 + 0.9066x + 0.0215$

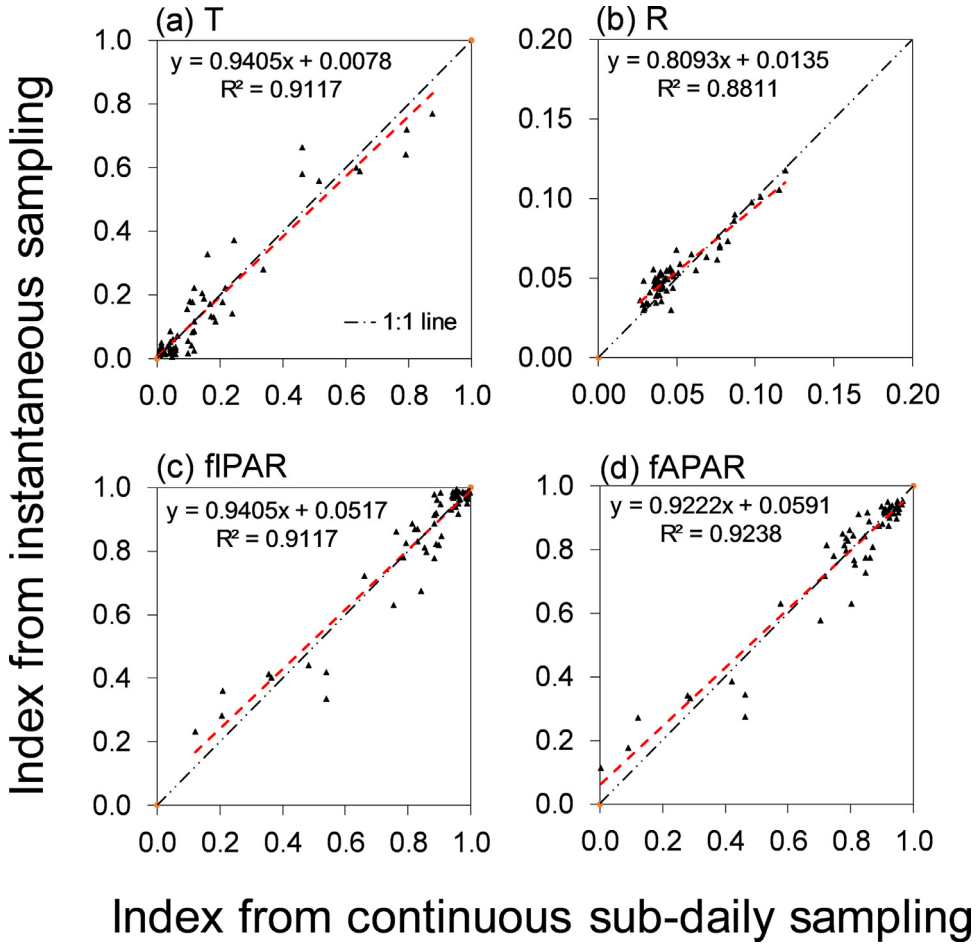


Fig. 14. Light interaction indices (a) Transmittance; (b) Reflectance; (c) fraction of intercepted photosynthetically active radiation (fIPAR); (d) fraction of absorbed photosynthetically active radiation (fAPAR) computed using instantaneous measurements using a ceptometer vs. true indices derived from aggregating sub-daily continuous samplings. Each index was subjected to a linear regression among the two sources of data.

indices, which are phases of canopy development. The RMSE between the solar noon sampling derived fAPAR and true fAPAR is 0.08 units, while the error was observed to be as high as 0.34 (110%). The similarity of the findings of the two analyses presented here implies that the solar-noon sampling, while may not present a bias when compared to true canopy light characteristics, it still can introduce substantial errors in the estimation of important indicators such as fAPAR, at least during the canopy development phase. These errors can consequently propagate into estimation of total light absorbed by the canopy during a period and hence, light use efficiency. Thus, although significant literature (Albrizio and Steduto, 2005; Mariscal et al., 2000; Edwards et al., 2005; Gou et al., 2017; Narayanan et al., 2013; Caviglia and Sadras, 2001) has used solar-noon measurement of light balance components, we have demonstrated that these might not be reflective of the true canopy light interactions, especially during canopy developmental stages, and hence should not be used as a proxy to high-frequency and continuous light balance measurements. The misrepresentation of the light interaction dynamics can be somewhat eliminated by adopting certain ceptometer deployment methods, as outlined by Johnson et al. (2010). We recommend that our findings on the potential misleading nature of instantaneously measured light interactions be acknowledged with additional sampling considerations suggested by Singer et al. (2011), to quantify light interception and absorption at the field-scale.

3.7. Light Extinction coefficient

Light extinction coefficient (k) based on Beer's law (Runyon et al., 1994; Sampson and Allen, 1998) was estimated at seasonal and sub-seasonal levels for each of the four crops. Fig. 16 presents the linear regression of ln(1-fIPAR) and LAI for each crop, the slope of which describes a seasonal estimate of k. These quantities had high correlations ($R^2 > 0.94$), except for winter wheat 2017–2018, which had relatively lower, but still high correlation ($R^2 = 0.83$). Specifically, estimates of k (listed in Table 2) were similar for S.S. maize and L.S. maize (0.45 and 0.46, respectively), while also being identical for soybean and sorghum (0.48 each), and were highly contrasting for the two winter wheat growing seasons (0.36 vs. 0.66 in 2016-2017 and 2017 vs. 2018 growing seasons). For this reason, the k was obtained separately for the two winter wheat growing seasons, unlike other crops, where the growing seasons were pooled due to their very similar scales of magnitude. Winter wheat in 2017-2018 growing season demonstrated a 50% reduction in LAI, owing to the abrupt increase in CTU around mid-growing season, although the light interception characteristics were similar, driving the k value higher than those in 2016-2017 growing season. Comparisons of k measured in this research have to consider that the corresponding LAI values are comparable as well for a given crop. As mentioned earlier, measured LAI in this research, especially for soybean, sorghum and winter wheat are generally higher than those reported in the literature, and consequently, k values for these crops are lower. Maize in high-yielding Nebraska conditions was evaluated for k by Lindquist et al. (2005), reporting pre-anthesis and post-anthesis k

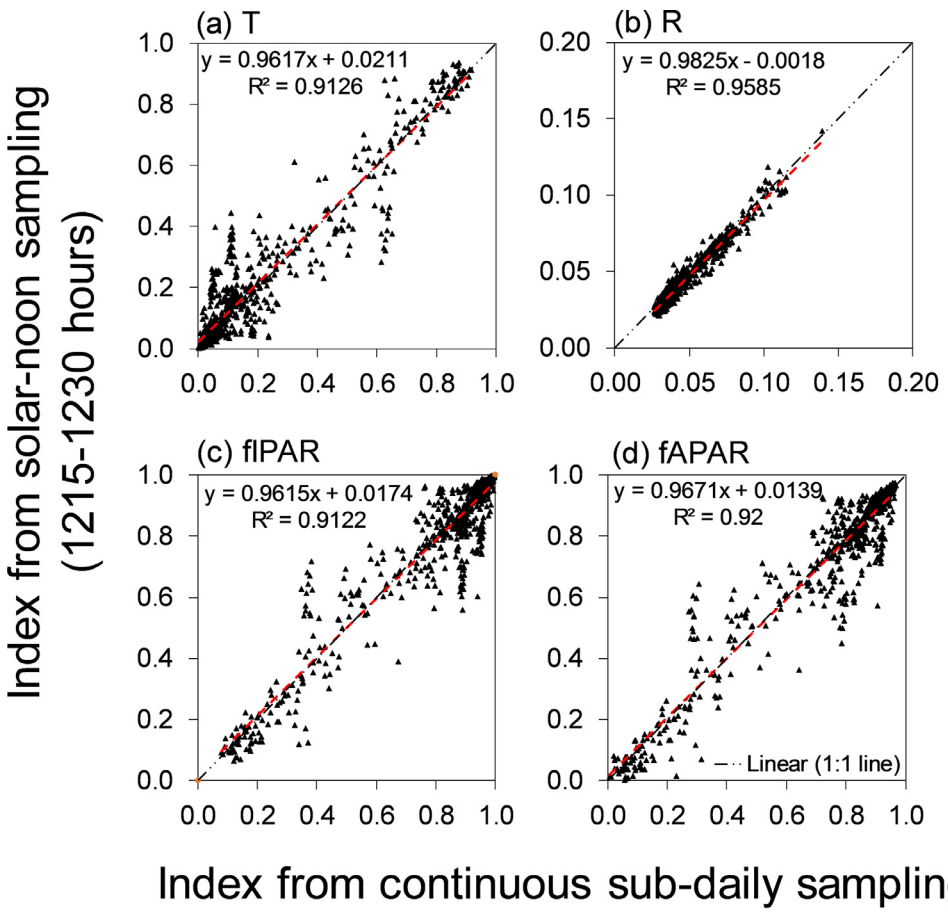


Fig. 15. Light interaction indices (a) Transmittance; (b) Reflectance; (c) fraction of intercepted photosynthetically active radiation (fIPAR); (d) fraction of absorbed photosynthetically active radiation (fAPAR) computed using solar noon sampling (1215–1230 h) vs. true indices derived from aggregating sub-daily continuous samplings. Each index was subjected to a linear regression among the two sources of data.

values to be 0.49 and 0.70. While their pre-anthesis k are similar to our results, the higher post-anthesis k result due to their use of green LAI, as opposed to our choice of total LAI, and hence should not be confused as contrasting findings.

The magnitudes of k were also studied sub-seasonally, by estimating k using the same procedure [k = -ln(T)/LAI] at several points during the growing seasons (Fig. 17). It was found that the mean value estimated at the seasonal level was subject to considerable variability (see CV values in Table 2) when sub-seasonal magnitudes were considered. In general, all crops showed increasing estimates of k as the growing season progressed and anywhere from 24 to 49% of the variability in k was explained by time (DAE). While all the crops demonstrated k values that were in the range of 0.2–0.6 (CV of 14–18%), winter wheat k ranged from 0.10-0.82 (CV of 35%). While comparing across crop canopies, k can be interpreted as the comparative estimate of light transmission into the canopy, relative to its morphological area (LAI). Thus, some canopies might allow more light transmission, at a given LAI than others, than any other given crop (as indicated earlier in Fig. 12), and consequently have greater k, implying that the relative rates of LAI and fAPAR govern the value of k. The non-static nature of k observed in this research signifies the sub-seasonal variability that occurs in canopy light interactions. This variability stems from changes with development stage, plant structure, canopy configurations (Evers et al., 2009) and architectural features including leaf shape and angle and internode length (Kahlen et al., 2008). As pointed out by Zhang et al. (2014), k is assumed as a constant in crop models, biogeochemical models, and remote sensing models that predict evapotranspiration, crop biomass, yield, or gross primary productivity. k is a primary factor that governs prediction of cropland and ecosystem carbon and water processes in these models, which has been demonstrated by several studies (White et al., 2000; Domingo et al., 1999; Brenner and Incoll, 1997) by conducting a sensitivity analysis of net primary production and evapotranspiration by incrementally changing k values, especially in the context of ecosystem and ecological modeling. This makes it important to have reliable estimates of k variability for a given vegetative surface under a given set of environmental conditions, which is what this research has achieved for maize, soybean, sorghum and winter wheat grown in semi-arid conditions.

3.7. Light use efficiency

Light use efficiency was calculated on a seasonal basis for each crop from measurements of aboveground biomass and cumulative absorbed PAR (Fig. 18). The estimations of LUE were made based on both absorbed and intercepted PAR (LUE_a and LUE_i, respectively), to be able to aid in fair comparisons across LUE values available in the literature (depending on the choice of intercepted or absorbed light). The LUE_a ranged across 1.8-fold magnitude across different crops and was the highest for L.S. maize, followed by S.S. maize, sorghum, winter wheat and soybean (Table 2). LUE_i for all the crops was expectedly lower than LUE_a for all crops (Table 2), given that the absorbed light is always a certain fraction of intercepted light. LUE_i spanned a similar range of crop-specific values; however, winter wheat had higher LUE_i than sorghum, which is in contrast to what was found when LUE_a was adopted as an LUE indicator. This behavior can be explained by differences in the reflectance characteristics of these crops, as LUE_i did not account for the higher reflectance, and hence lower fAPAR, and consequently, higher LUE_a in sorghum. This stresses the need for careful consideration while selection of LUE basis, as LUE_a represents a more realistic representation of the efficiency of a canopy to make use of available light. We have already demonstrated that although R is a relatively smaller variable magnitude-wise, but can dictate the canopy

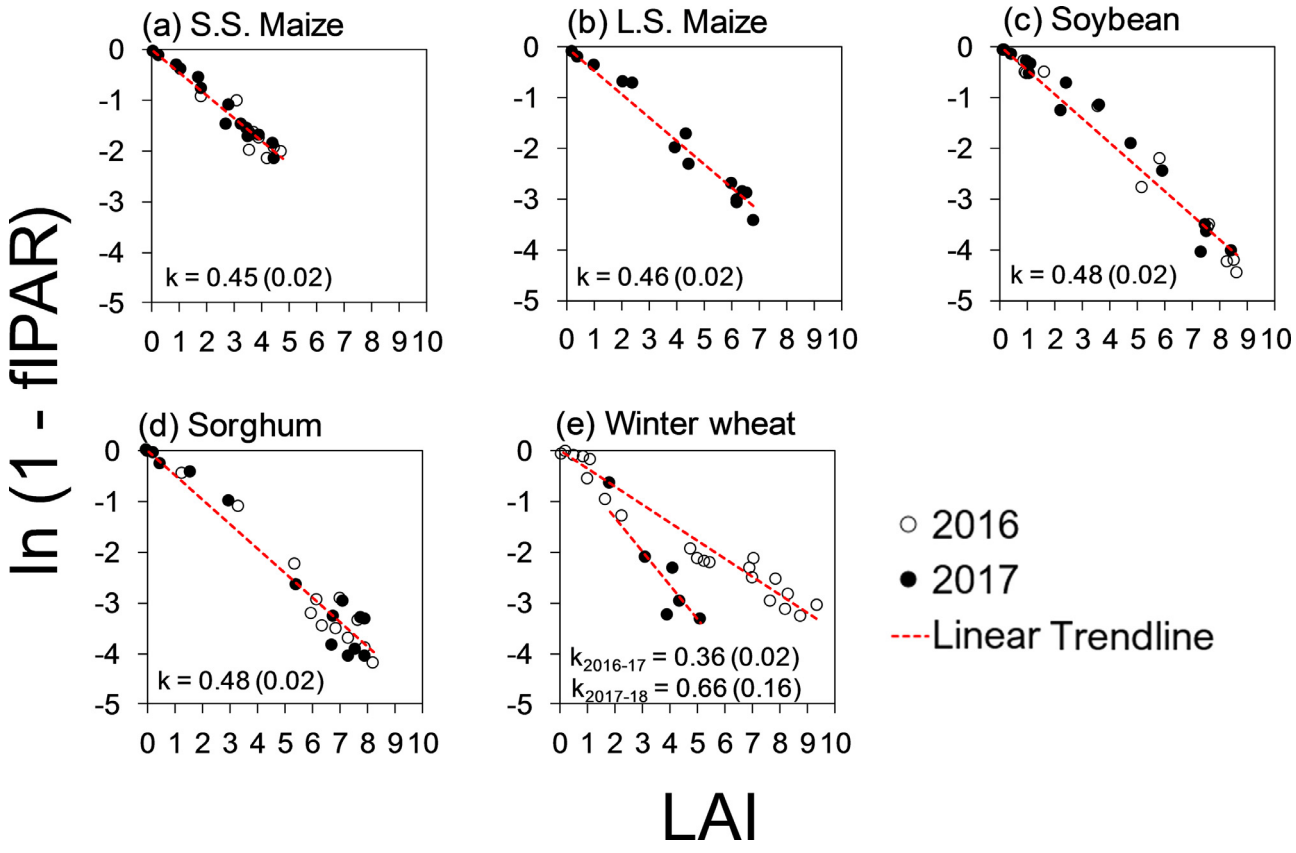


Fig. 16. Seasonal mean magnitudes of light extinction coefficients (k) derived as the slope of a linear regression among the two quantities, i.e., ln(1-fIPAR) and LAI for (a) S.S. maize; (b) L.S. maize; (c) soybean; (d) sorghum; (e) winter wheat. The value in the parenthesis shows the standard error associated with the k estimates.

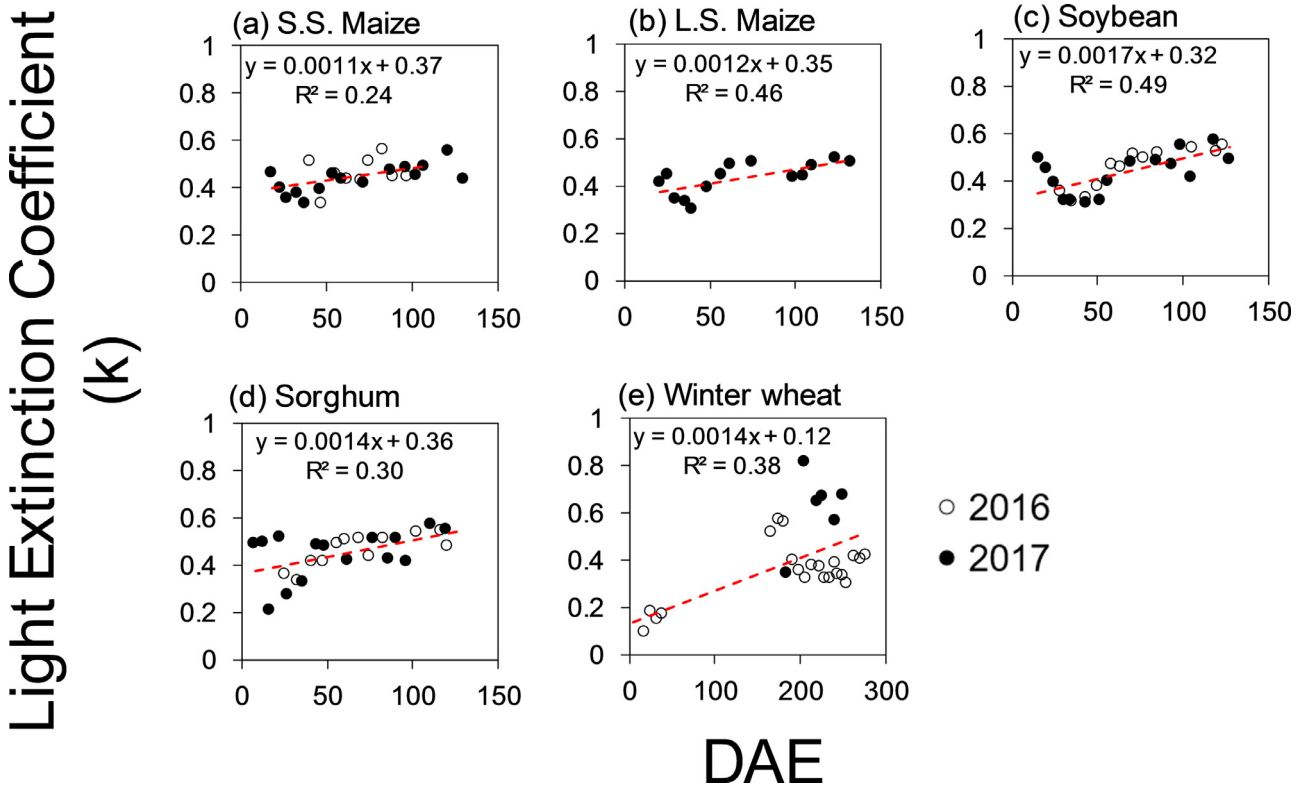


Fig. 17. The sub-seasonal variability in light extinction coefficients (k) measured for several short intervals during the two growing seasons for (a) S.S. maize; (b) L.S. maize; (c) soybean; (d) sorghum; (e) winter wheat in relation to days after emergence.

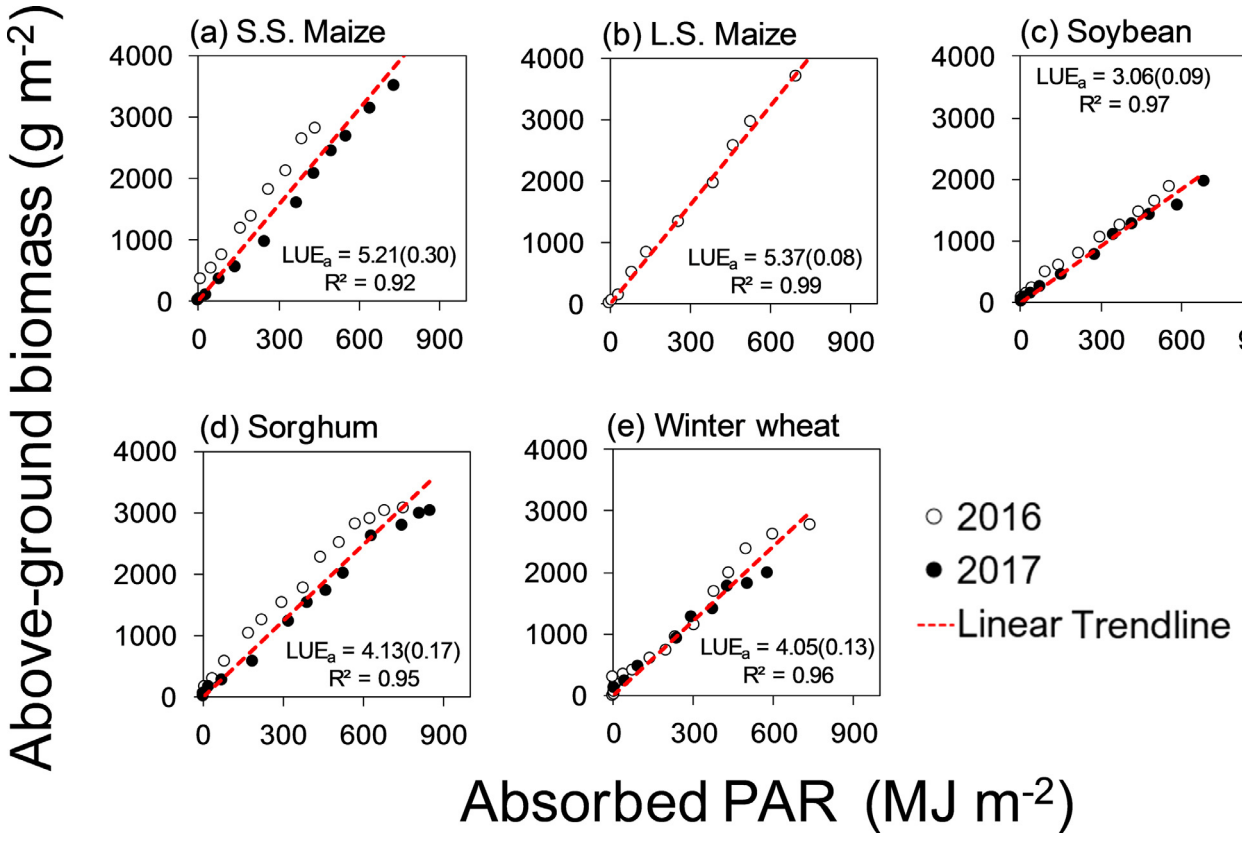


Fig. 18. Light use efficiency based on absorbed PAR (LUE_a) estimated as the slope of linear regression among aboveground biomass (AGB) and cumulative absorbed PAR for (a) S.S. maize; (b) L.S. maize; (c) soybean; (d) sorghum; (e) winter wheat. The value in the parenthesis shows the standard error associated with the LUE_a estimates.

light use patterns to a larger extent, in certain phases of growth. The estimates of LUE estimated in this research, are generally greater than that reported in the literature. For example, previous LUE research conducted in optimal growth conditions in Nebraska (Lindquist et al., 2005) reported maize LUE as 3.84 g MJ^{-1} APAR, or 3.73 g MJ^{-1} IPAR, respectively, which are 26 and 23% lower than our APAR and IPAR based estimates for S.S. maize, respectively, and 28 and 27% lower than our APAR and IPAR based estimates for L.S. maize, respectively. Previous LUE estimates for proximal environmental conditions (Nebraska) do not exist for the rest of the four crops, but research at other sites addressing LUE in maize (Singer et al., 2011; Hatfield 2014; Edwards et al., 2005), soybean (Singer et al., 2011; Purcell et al., 2002; Hatfield 2014), sorghum (Albrizio et al., 2005; Ceotto et al., 2013; Van Roekel and Purcell, 2014) and winter wheat (Albrizio et al., 2005; O'Connell et al., 2004; Caviglia and Sadras, 2001) also reported lower LUE values than in this research. Through this research, we want to suggest that these current LUE measured at SCAL, Nebraska, represent the values for these four major cropping systems grown under optimal growth conditions and may also represent the upper limit (or close to it) of LUE values.

LUE estimates were also quantified for shorter sub-seasonal periods (corresponding to AGB samplings) for each crop to decipher the variability associated with mean LUE estimates. These were computed using incremental aboveground biomass and light absorbed since previous consecutive sampling and are presented in Fig. 19. We find that each crop was subject to substantial variability in LUE values, which ranged from a minimum of 1.02 g MJ^{-1} [standard deviation (SD) (or a coefficient of variation (CV] of 33%) in soybean to a maximum of 2.17 g MJ^{-1} (SD) (or a CV of 54%) in winter wheat. The crop-specific SD and CV associated with sub-seasonal LUE_a are provided in Table 2. This signifies the fact that LUE is scale-dependent and can vary largely, depending on several factors, including crop phenological stages and environmental conditions such as vapor pressure deficit (Stockle and Kiniry, 1990; Kiniry et al., 1998; Sinclair and Muchow, 1999; Kemanian et al., 2004) and soil-water and plant water status as well as soil and plant nutrient status, which in turn, influence plant turgidity and leaf expansion, which influence LAI and LUE, under a given set of soil and crop management practices, which were maintained constant in our research framework. For example, the higher variability observed in winter wheat LUE_a is likely due to the wide range of weather conditions encapsulated within its growing season (October until July), which was largest than the rest of the crops. While LUE variability has been addressed under several cropping systems at contrasting sites and cropping seasons (Garbulsky et al., 2010), but evidence of within-season LUE variability is limited (irrigated rice crop studied by Campbell et al., 2001) and this research has quantified the extent of variability in crop-specific LUE using field-measured data.

Despite the misgivings regarding the use of LUE_i to characterize crop canopies, it would be ideal for any research in this direction to present estimates of both LUE_a and LUE_i, in order to have a fair comparison across prior research, under different regimes of growth conditions. In the cases where the means to measure reflectance do not exist, empirical relationships between the two can be used to estimate LUE_a from LUE_i, directly. This approach can also be extended to other LUE_i values reported in the literature, provided that the relationships are unaffected by variation in management, soils, stresses, climate, etc. Using a linear regression at seasonal and sub-seasonal scales, it was found that LUE_a was 5 and 7% greater than LUE_i, respectively (Fig. 20(a)). To determine if the relative magnitudes of LUE_a and LUE_i was affected during the growing season, the ratio was evaluated across the growing season using a normalized cumulative thermal unit (NCTU) as the base scale (Fig. 17b). NCTU was computed as the CTU at a given

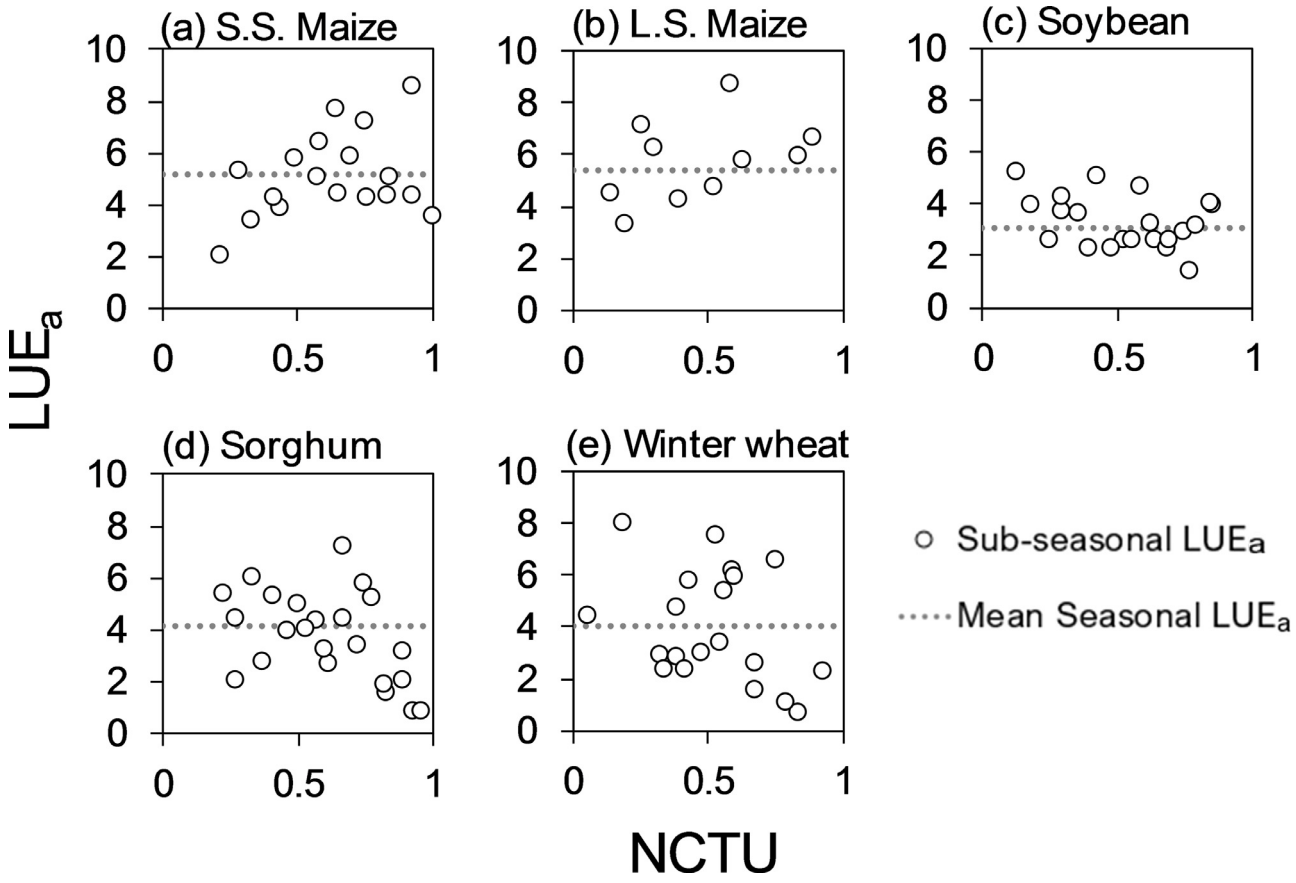


Fig. 19. The sub-seasonal variability in light use efficiency based on absorbed PAR (LUE_a) measured for several short intervals during the two growing seasons for (a) S.S. maize; (b) L.S. maize; (c) soybean; (d) sorghum; (e) winter wheat in relation to normalized cumulative thermal units (NCTU).

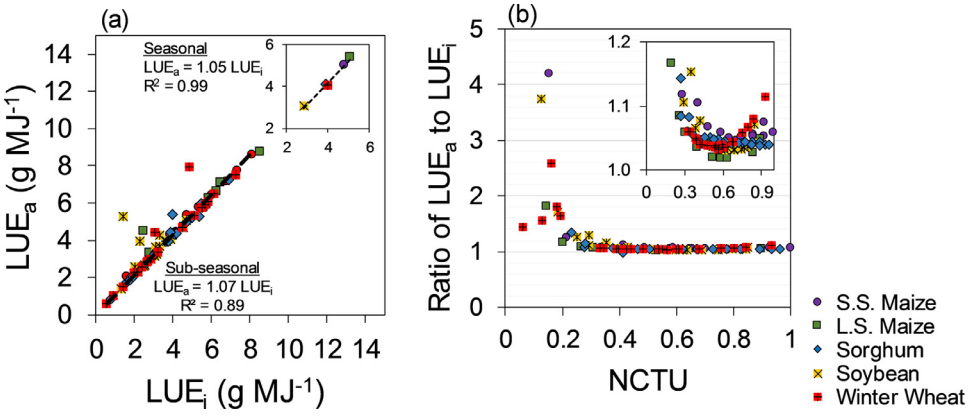


Fig. 20. (a) Linear regression among light use efficiency based on absorbed PAR (LUE_a) and light use efficiency based on intercepted PAR (LUE_i) for sub-seasonal intervals for each crop. The intercepted was forced to origin. The inset figure shows the same linear regression analysis, but for seasonal mean LUE estimates for each crop. (b) The ratio of LUEa to LUEi in relation to normalized cumulative thermal units (NCTU) for each crop. The inset figure shows a zoomed version with the Y-axes restricted to 1–1.2.

time to the CTU at harvest, and hence the growing season for the two seasons and each crop was normalized to a scale of 0 to 1. These analyses revealed that the differences in LUE_a and LUE_i are a function of growth stage. Earlier in the growing season, higher ratios were observed (as high as 4.2), which then abruptly declined to a range of 1 to 2 at around 0.2 NCTU (20% completion of the crop duration) until the end of the crop duration. The inset of Fig. 20(b) shows the same curve with Y-axis restricted to 1 and 1.2 and shows the growing season variation of the ratio, which was undetectable otherwise owing to scale. The ratio roughly demonstrates a U-shaped curve, characterized by lowest ratio value around mid-crop duration, and higher ratios (approaching 1.1-1.2) at around 0.2 and 1.0 NCTU. This variation stems from the seasonal variation in R, which was unaccounted for in LUE_i, unlike LUE_a, as explained earlier. Thus, since majority of the growing season represent smaller ratios (1.02-1.10), this conversion factor can be used, provided NCTU > 0.25. A similar factor of 0.972 was used by Lindquist et al. (2005) to convert LUE based on APAR to that based on IPAR. Despite having low magnitudes, R can be a fairly impactful component of the light balance. The robust fIPAR vs. fAPAR, and LUE_i vs LUE_a relationships presented here can aid in accurately accounting for canopy light interactions without measuring R. Use of these relationships can avoid additional cost of instrumentation for measuring R, while still maintaining data integrity and accuracy.

4. Conclusions

In this research, light fluxes were measured in four major row crops in the U.S. High Plains region, i.e., maize, soybean, sorghum and winter

wheat during 2016–2018. Measurements followed two complementary approaches that differed in their frequency and temporal representativeness of light dynamics. Experiments conducted in this research aimed to provide optimum growing conditions for near-genetic potential productivity, using well-informed decision-making in terms of water and nitrogen, in order to represent the high productivity of High Plains region production systems. Several indices that reflect and convey light interaction, light use, and light use efficiency of these cropping systems were quantified/estimated, that included transmittance (R), reflectance (R), fraction of intercepted PAR (fIPAR), fraction of absorbed PAR (fAPAR), light extinction coefficients (k), light use efficiency (LUE) based on intercepted (LUE_i) and absorbed light (LUE_a). Every crop was demonstrated to present varied signatures in their light interaction patterns, both at seasonal and sub-seasonal scales, and these crop-specific characteristics were presented graphically and quantitatively.

Due to the management practices aimed at input sufficiency, direct and indirect crop productivity indicators were shown to communicate higher productivity levels. This included higher leaf areas index (LAI), aboveground biomass (AGB), and light use efficiency (LUE), and lower extinction coefficients (k), in comparison to the values found in the literature for these crops. Some of these metrics are extensively used as important parameters in crop models to simulate crop dry matter accumulation and inaccurate calibration of these variables can substantially influence the model simulation performance and modeling outcomes and related decision-making. The research also provides insights into the potential drawbacks associated with light sampling and reporting approaches that could lead to misrepresentation and misleading interpretation of light interaction dynamics in a given canopy. A detailed investigation on the light balance sampling approach in this research, i.e., ceptometer measurements around solar noon vs. continuous light balance sampling yielded insights into the trade-offs among the two approaches. While there is significant cost incurred with employment of continuous light sampling infrastructure (Salter et al., 2019), especially when canopy non-uniformity necessitates multiple monitoring units (Singer et al., 2011), it provides shielding against non-representative estimates of true light interactions that can occur during incomplete canopy closure when light sampling is conducted at or around solar noon. Thus, significant caution should be used when upscaling light interactions recorded instantaneously to represent mean daily interactions. Our research also investigated relative magnitudes and transferability between fIPAR and fAPAR, and consequently LUE based on the latter and former, and found that the multiplicative factors suggested by previous research (Bonhomme, 2000; Sinclair and Muchow, 1999) underestimate by around 12% for crops grown at the research site under optimal management conditions. Moreover, empirical models were developed to aid in efficient conversion from light interception and absorption and proposed for future use.

Conclusively, this research serves as the sole assessment of comparative light interaction metrics across major U.S. row crops under the current levels of productivity achieved. Measured data on light use parameters in the research is of value to a range of crop modeling applications in light, water and energy balance for the row crop agricultural landscapes.

Declaration of Competing Interests

The authors declare no conflict of interest.

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References

Albrizio, R., Steduto, P., 2005. Resource use efficiency of field-grown sunflower, sorghum, wheat and chickpea: I. Radiat. effic. Agric. For. Meteorol. 130, 254–268.
Bingham, I.J., Blake, J., Foulkes, M.J., Spink, J., 2007. Is barley yield in the UK sink limited?: I. Post-anthesis radiation interception, radiation-use efficiency and source–sink balance. Field Crops Res 101, 198–211.
Blackman, V.H., 1919. The compound interest law and plant growth. Ann. Botany 33, 353–360.
Bonhomme, R., 2000. Beware of comparing RUE values calculated from PAR vs solar radiation or absorbed vs intercepted radiation. Field Crops Res 68, 247–252.
Brenner, A.J., Incoll, L.D., 1997. The effect of clumping and stomatal response on evaporation from sparsely vegetated shrublands. Agric. For. Meteorol. 84, 187–205.
Brisson, N., Gary, C., Justes, E., Roche, R., Mary, B., Ripoche, D., Zimmer, D., Sierra, J., Bertuzzi, P., Burger, P., 2003. An overview of the crop model STICS. Eur. J. Agron. 18, 309–332.
Camargo, D.C., Montoya, F., Moreno, M.A., Ortega, J.F., Corcoles, J.I., 2016. Impact of water deficit on light interception, radiation use efficiency and leaf area index in a potato crop (Solanum tuberosum L.). J. Agric. Sci. 154, 662–673.
Campbell, C.S., Heilman, J.L., McInnes, K.J., Wilson, L.T., Medley, J.C., Wu, G., Cobos, D.R., 2001. Seasonal variation in radiation use efficiency of irrigated rice. Agric. For. Meteorol. 110, 45–54.
Caviglia, O.P., Sadras, V.O., 2001. Effect of nitrogen supply on crop conductance, water- and radiation-use efficiency of wheat. Field Crops Res 69, 259–266.
Ceotto, E., Di Candilo, M., Castelli, F., Badeck, F., Rizza, F., Soave, C., Volta, A., Villani, G., Marletto, V., 2013. Comparing solar radiation interception and use efficiency for the energy crops giant reed (Arundo donax L.) and sweet sorghum (Sorghum bicolor L. Moench). Field Crops Res. 149, 159–166.
De Wit, C.T., 1959. Potential photosynthesis crop surfaces. Neth. J. Agric. Sci. 7, 141–149.
Domingo, F., Villagarcıa, L., Brenner, A.J., Puigdefábregas, J., 1999. Evapotranspiration model for semi-arid shrub-lands tested against data from SE Spain. Agric. For. Meteorol. 95, 67–84.
Edwards, J.T., Purcell, L.C., Vories, E.D., 2005. Light interception and yield potential of short-season maize (Zea mays L.) hybrids in the Midsouth. Agron. J. 97, 225–234.
Evers, J.B., Huth, N.I., Renton, M., 2009. Light extinction in spring wheat canopies in relation to crop configuration and solar angle. 2009 plant growth modeling and applications. IEEE 107.
Fletcher, A.L., Johnstone, P.R., Chakwizira, E., Brown, H.E., 2013. Radiation capture and radiation use efficiency in response to N supply for crop species with contrasting canopies. Field Crops Res 150, 126–134.
Gao, Y., Duan, A., Qiu, X., Sun, J., Zhang, J., Liu, H., Wang, H., 2010. Distribution and use efficiency of photosynthetically active radiation in strip intercropping of maize and soybean. Agron. J. 102, 1149–1157.
Garbulsky, M.F., Peñuelas, J., Papale, D., Ardö, J., Goulden, M.L., Kiely, G., Richardson, A.D., Rotenberg, E., Veenendaal, E.M., Filella, I., 2010. Patterns and controls of the variability of radiation use efficiency and primary productivity across terrestrial ecosystems. Global Ecol. Biogeogr. 19, 253–267.
Gou, F., van Ittersum, M.K., Simon, E., Leffelaar, P.A., van der Putten, Peter, EL, Zhang, L., van der Werf, W., 2017. Intercropping wheat and maize increases total radiation interception and wheat RUE but lowers maize RUE. Eur. J. Agron. 84, 125–139.
Guiducci, M., Antognoni, A., Benincasa, P., 1992. Effect of water availability on leaf movement, light interception and light utilisation efficiency in several field crops. Ri ista di Agronomia 27, 392–397.
Hatfield, J.L., 2014. Radiation use efficiency: Evaluation of cropping and management systems. Agron. J. 106, 1820–1827.
Irmak, S., 2010. Nebraska water and energy flux measurement, modeling, and research network (NEBFLUX). Trans. ASABE 53 (4), 1097–1115.
Irmak, S., Djaman, K., 2016. Effects of planting date and density on plant growth, yield, evapotranspiration, and water productivity of subsurface drip-irrigated and rainfed maize. Trans. ASABE 59, 1235–1256.
Irmak, S., Mutiibwa, D., 2008. Dynamics of photosynthetic photon flux density and light extinction coefficient for assessing radiant energy interactions for maize canopy. Trans. ASABE 51 (5), 1663–1673.
Jacovides, C.P., Timbios, F., Asimakopoulos, D.N., Steven, M.D., 1997. Urban aerosol and clear skies spectra for global and diffuse photosynthetically active radiation. Agric. For. Meteorol. 87, 91–104.
Johnson, M.V., Kiniry, J.R., Burson, B.L., 2010. Ceptometer deployment method affects measurement of fraction of intercepted photosynthetically active radiation. Agron. J. 102, 1132–1137.
Jones, C.A., Dyke, P.T., Williams, J.R., Kiniry, J.R., Benson, V.W., Griggs, R.H., 1991. EPIC: an operational model for evaluation of agricultural sustainability. Agricult. Syst. 37, 341–350.
Jones CA, Kiniry JR, Dyke PT. 1986. CERES-maize: A simulation model of maize growth and development. TexasA& M University Press.
Jones, J.W., Hoogenboom, G., Porter, C.H., Boote, K.J., Batchelor, W.D., Hunt, L.A., Ritchie, J.T., 2003. The DSSAT cropping system model. Eur. J. of Agron. 18 (3–4), 235–265.
Kahlen, K., Wiechers, D., Stützel, H., 2008. Modelling leaf phototropism in a cucumber canopy. Funct. Plant Biol. 35, 876–884.
Kemanian, A.R., Stöckle, C.O., Huggins, D.R., 2004. Variability of barley radiation-use efficiency. Crop Sci 44, 1662–1672.
Kiniry, J.R., Simpson, C.E., Schubert, A.M., Reed, J.D., 2005. Peanut leaf area index, light interception, radiation use efficiency, and harvest index at three sites in Texas. Field Crops Res 91, 297–306.
Kiniry, J.R., Landivar, J.A., Witt, M., Gerik, T.J., Cavero, J., Wade, L.J., 1998. Radiation-

use efficiency response to vapor pressure deficit for maize and sorghum. Field Crops Res 56, 265–270.
Kiniry, J.R., Jones, C.A., O'toole, J.C., Blanchet, R., Cabelguenne, M., Spanel, D.A., 1989. Radiation-use efficiency in biomass accumulation prior to grain-filling for five grain-crop species. Field Crops Res. 20, 51–64.
Kukal, M.S., Irmak, S., 2018. Climate-driven crop yield and yield variability and climate change impacts on the US great plains agricultural production. Scient. Rep. 8, 3450.
Kukal, M.S., Irmak, S., 2018. US Agro-Climate in 20 th Century: Growing Degree Days, First and Last Frost, Growing Season Length, and Impacts on Crop Yields. Scient. Rep. 8, 6977.
Kukal, M.S., Irmak, S., 2019. Comparative canopy growth dynamics in four row crops and their relationships with allometric and environmental determinants. Agron J 111, 1799–1816.
Kukal, M.S., Irmak, S., 2020. Characterization of water use and productivity dynamics across four C3 and C4 row crops under optimal growth conditions. Agric. Water Man. 227, 105840.
Lecoeur, J., Ney, B., 2003. Change with time in potential radiation-use efficiency in field pea. Eur. J. Agron. 19, 91–105.
Li, Q., Chen, Y., Liu, M., Zhou, X., Yu, S., Dong, B., 2008. Effects of irrigation and planting patterns on radiation use efficiency and yield of winter wheat in North China. Agric. Water Manage. 95, 469–476.
Lindquist, J.L., Arkebauer, T.J., Walters, D.T., Cassman, K.G., Dobermann, A., 2005. Maize radiation use efficiency under optimal growth conditions. Agron. J. 97, 72–78.
Loomis, R.S., Williams, W.A., 1963. Maximum crop productivity: an extimate 1. Crop Sci 3, 67–72.
Maddonni, G.A., Otegui, M.E., 1996. Leaf area, light interception, and crop development in maize. Field Crops Res 48, 81.
Mariscal, M.J., Orgaz, F., Villalobos, F.J., 2000. Radiation-use efficiency and dry matter partitioning of a young olive (Olea europaea) orchard. Tree Physiol 20, 65–72.
McCartney, H.A., 1978. Spectral distribution of solar radiation II: Global and diffuse. Q. J. R. Meteorol. Soc. 104, 911–926.
McCree, K.J., 1972. Test of current definitions of photosynthetically active radiation against leaf photosynthesis data. Agric. Meteorol. 10, 443–453.
McMaster, G.S., Wilhelm, W.W., 1997. Growing degree-days: one equation, two interpretations. Agric. For. Meteorol. 87, 291–300.
Monteith, J.L., 1972. Solar radiation and productivity in tropical ecosystems. J. Appl. Ecol. 9, 747–766.
Monteith, J.L., 1977. Climate and the efficiency of crop production in Britain. Phil. Trans. R. Soc. Lond. B. 281, 277–294.
Muchow, R.C., Sinclair, T.R., Bennett, J.M., 1990. Temperature and solar radiation effects on potential maize yield across locations. Agron. J. 82, 338–343.
Narayanan, S., Aiken, R.M., Vara Prasad, P.V., Xin, Z., Yu, J., 2013. Water and radiation use efficiencies in sorghum. Agron. J. 105, 649–656.
NASS, U., 2018. Census of agriculture. US Department of Agriculture, National Agricultural Statistics Service, Washington, DC, pp. 1.
NASS, U., 2018. United States Department of Agriculture-National Agricultural Statistics Service. Cropland Data Layer Accessed on October 5 th, 2018.
O'Connell, M.G., O'Leary, G.J., Whitfield, D.M., Connor, D.J., 2004. Interception of photosynthetically active radiation and radiation-use efficiency of wheat, field pea and mustard in a semi-arid environment. Field Crops Res. 85, 111–124.
Plénet, D., Mollier, A., Pellerin, S., 2000. Growth analysis of maize field crops under phosphorus deficiency. II. Radiation-use efficiency, biomass accumulation and yield components. Plant Soil. 224, 259–272.
Pradhan, S., Sehgal, V.K., Das, D.K., Jain, A.K., Bandyopadhyay, K.K., Singh, R., Sharma, P.K., 2014. Effect of weather on seed yield and radiation and water use efficiency of mustard cultivars in a semi-arid environment. Agric. Water Manage. 139, 43–52.
Purcell, L.C., Ball, R.A., Reaper, J.D., Vories, E.D., 2002. Radiation use efficiency and biomass production in soybean at different plant population densities. Crop Sci 42, 172–177.
Rochette, P., Desjardins, R.L., Pattey, E., Lessard, R., 1996. Instantaneous measurement of radiation and water use efficiencies of a maize crop. Agron. J. 88, 627–635.
Ruiz, R.A., Bertero, H.D., 2008. Light interception and radiation use efficiency in temperate quinoa (Chenopodium quinoa Willd.) cultivars. Eur. J. Agron. 29, 144–152.
Runyon, J., Waring, R.H., Goward, S.N., Welles, J.M., 1994. Environmental limits on net primary production and light-use efficiency across the Oregon transect. Ecol. Appl. 4, 226–237.
Russell, G., Jarvis, P.G., Monteith, J.L., 1989. Absorption of radiation by canopies and stand growth. In: Russell, G., Marshall, B., Jarvis, P.G. (Eds.), Plant Canopies: Their Growth, Form and Function. Cambridge University Press, Cambridge, pp. 21–39.
Salter, W.T., Merchant, A.M., Gilbert, M.E., Buckley, T.N., 2019. PARbars: cheap, easy to build ceptometers for continuous measurement of light interception in plant canopies. JoVE (J. Visual. Exper.) 147, e59447.
Sampson, D.A., Allen, H.L., 1998. Light attenuation in a 14-year-old loblolly pine stand as influenced by fertilization and irrigation. Trees 13, 80–87.
Sinclair, TR, Muchow, RC, 1999. Radiation use efficiency. Adv. in Agron 65, 215–265.
Singer, J.W., Meek, D.W., Sauer, T.J., Prueger, J.H., Hatfield, J.L., 2011. Variability of light interception and radiation use efficiency in maize and soybean. Field Crops Res 121, 147–152.
Steduto, P., Todorovic, M., Caliandro, A., Rubino, P., 2003. Daily reference evapotranspiration estimates by the Penman-Monteith equation in Southern Italy. Constant vs. variable canopy resistance. Theor. and Appl. Clim. 74, 217–225.
Steduto, P., Albrizio, R., 2005. Resource use efficiency of field-grown sunflower, sorghum, wheat and chickpea: II. Water use efficiency and comparison with radiation use efficiency. Agric. For. Meteorol. 130, 269–281.
Stöckle, C.O., Kiniry, J.R., 1990. Variability in crop radiation-use efficiency associated with vapor-pressure deficit. Field Crops Res 25, 171–181.
Stöckle, C.O., Donatelli, M., Nelson, R., 2003. CropSyst, a cropping systems simulation model. Eur. J. Agron. 18, 289–307.
Tei, F., Scaife, A., Aikman, D.P., 1996. Growth of lettuce, onion, and red beet. 1. Growth analysis, light interception, and radiation use efficiency. Ann. Botany 78, 633–643.
Teixeira, E.I., George, M., Herreman, T., Brown, H., Fletcher, A., Chakwizira, E., de Ruiter, J., Maley, S., Noble, A., 2014. The impact of water and nitrogen limitation on maize biomass and resource-use efficiencies for radiation, water and nitrogen. Field Crops Res 168, 109–118.
Tollenaar, M., Aguilera, A., 1992. Radiation use efficiency of an old and a new maize hybrid. Agron. J. 84, 536–541.
Van Roekel, R.J., Purcell, L.C., 2014. Soybean biomass and nitrogen accumulation rates and radiation use efficiency in a maximum yield environment. Crop Sci 54, 1189–1196.
Vieira, M.I., de Melo-Abreu, J.P., Ferreira, M.E., Monteiro, A.A., 2009. Dry matter and area partitioning, radiation interception and radiation-use efficiency in open-field bell pepper. Scient. Horticult. 121, 404–409.
Walulu, R.S., Rosenow, D.T., Wester, D.B., Nguyen, H.T., 1994. Inheritance of the stay green trait in sorghum. Crop Sci 34, 970–972.
Wang, J., Zhao, T., Wang, E., Yu, Q., Yang, X., Feng, L., Pan, X., 2010. Measurement and simulation of diurnal variations in water use efficiency and radiation use efficiency in an irrigated wheat-maize field in the North China Plain. N. Z. J. Crop Hortic. Sci. 38, 119–135.
Watson, DJ., 1952. The physiological basis of variation in yield. Advances in Agronomy 4, 101–145.
Weiss, M., Baret, F., Smith, G.J., Jonckheere, I., Coppin, P., 2004. Review of methods for in situ leaf area index (LAI) determination: Part II. Estimation of LAI, errors and sampling. Agric. For. Meteorol 121, 37–53.
White, M.A., Thornton, P.E., Running, S.W., Nemani, R.R., 2000. Parameterization and sensitivity analysis of the BIOME–BGC terrestrial ecosystem model: net primary production controls. Earth Interact 4, 1–85.
Zhang, L., Hu, Z., Fan, J., Zhou, D., Tang, F., 2014. A meta-analysis of the canopy light extinction coefficient in terrestrial ecosystems. Front. Earth Sci. 8, 599–609 Tables.